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CRISPR-GPT for agentic automation of gene-editing experiments

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Performing effective gene-editing experiments requires a deep understanding of both the CRISPR technology and the biological system involved. Meanwhile, despite their versatility and promise, large language models (LLMs) often lack domain-specific knowledge and struggle to accurately solve biological design problems. We present CRISPR-GPT, an LLM agent system to automate and enhance CRISPR-based gene-editing design and data analysis. CRISPR-GPT leverages the reasoning capabilities of LLMs for complex task decomposition, decision-making and interactive human–artificial intelligence (AI) collaboration. This system incorporates domain expertise, retrieval techniques, external tools and a specialized LLM fine tuned with open-forum discussions among scientists. CRISPR-GPT assists users in selecting CRISPR systems, experiment planning, designing guide RNAs, choosing delivery methods, drafting protocols, designing assays and analysing data. We showcase the potential of CRISPR-GPT by knocking out four genes with CRISPR-Cas12a in a human lung adenocarcinoma cell line and epigenetically activating two genes using CRISPR-dCas9 in a human melanoma cell line. CRISPR-GPT enables fully AI-guided gene-editing experiment design and analysis across different modalities, validating its effectiveness as an AI co-pilot in genome engineering.

Large language models (LLMs) have demonstrated exceptional capabilities in language skills and encapsulate a substantial amount of world knowledge[1–5]. Recent research has also enhanced LLMs with external tools, improving their problem-solving abilities and efficiencies[6–8]. Moreover, LLMs have also demonstrated potential as tool makers[9] and black-box optimizers[10]. To this end, researchers have explored LLM-based specialized models for various scientific domains[11,12], particularly for mathematics and chemistry tasks. ChemCrow[13] uses tool-augmented LLM for solving a range of chemistry-related tasks such as paracetamol synthesis, whereas Co-scientist[14] integrates automated experimentation, achieving successful optimization of palladium-catalysed cross-coupling reaction. LLMs have also shown initial promise in generating biological protocols, as demonstrated by studies like BioPlanner[15]. While recent advancements, such as OpenAI's o1 preview, have improved reasoning abilities in areas such as mathematics and coding, progress in biological tasks remains comparatively limited. This limitation stems from general-purpose LLMs' lack of in-depth understanding of biology,

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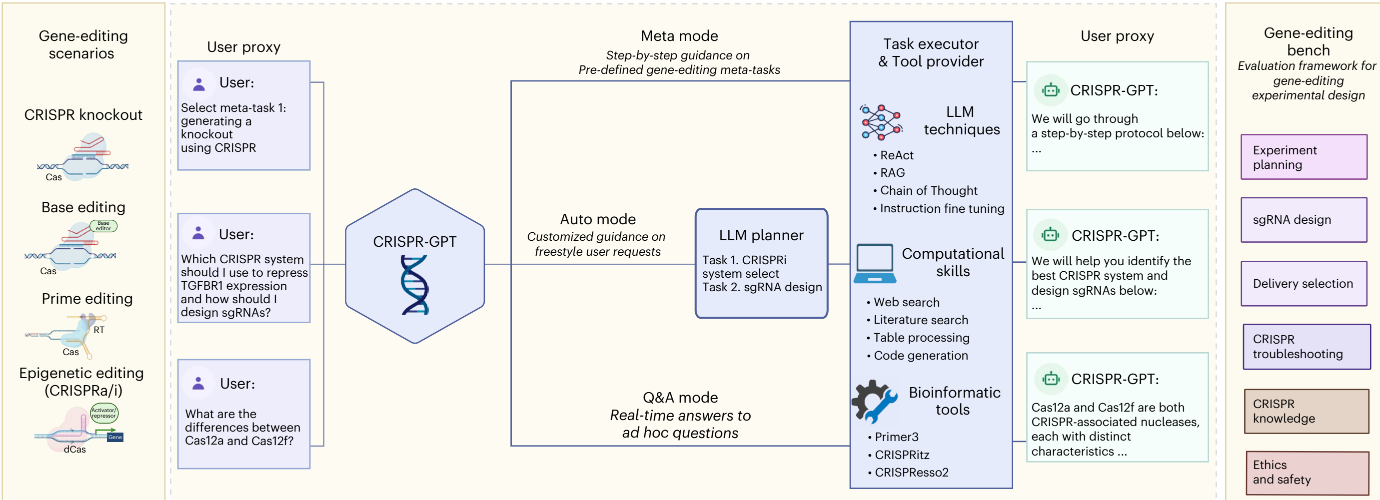

Fig. 1 | Overview of CRISPR-GPT. CRISPR-GPT is an LLM-powered multi-agent system designed to provide AI copiloting for human researchers in gene editing. It supports four primary gene-editing modalities: knockout, base editing, prime editing and epigenetic editing (CRISPRa/i). The system offers three user interaction modes: Meta mode (step-by-step guidance on predefined tasks), Auto mode (customized guidance based on user requests) and Q&A mode (real-time answers to ad hoc questions), to streamline experiment design and planning. CRISPR-GPT consists of four core components: the User proxy, LLM planner, Task executor and Tool provider. Together, these components are equipped with a comprehensive suite of tools and decision-support capabilities to facilitate the design, planning and analysis of gene-editing workflows. To evaluate CRISPR-GPT's performance, we developed the Gene-editing bench, a framework of 288 test cases covering tasks such as experimental planning, sgRNA design, delivery method selection and more. Figure was originally created with BioRender.com/tb8sq6f.

compounded by the unique challenges of biological experiments, including the variability of living systems, the noisy nature of biological data and the highly specialized, less transferable nature of biological skills and tools.

Gene editing has transformed biological research and medicine, allowing for precise DNA modifications for both therapeutic and experimental applications. CRISPR-Cas, the most well-known gene-editing technology, originated from bacterial immune systems[16–24]. Its development has led to advanced techniques like CRISPR activation and interference (CRISPRa/i)[25–29], base editing[30,31] and prime editing[32,33], creating a powerful toolkit for genetic modification and epigenetic modulation. In basic biomedical research, CRISPR gene-editing has become one of the most frequently used laboratory techniques: at the largest non-profit plasmid DNA repository, Addgene, 8 of the 15 top requested plasmids worldwide were for CRISPR gene-editing[34]. On the application side, CRISPR has produced the first permanent cure for sickle cell disease (SCD)[35] and β-thalassaemia[36], as well as facilitating plant engineering for sustainable agriculture[20]. As one of the most powerful biotechnologies, numerous software and protocols exist for specific gene-editing tasks. Despite these resources, an end-to-end solution—from CRISPR-Cas system selection, guide (g)RNA design, off-target evaluation, to delivery and data analysis—remains complex, particularly for newcomers. AI-assisted tools can simplify gene-editing experiment design and data analysis, making the technology more accessible and accelerating scientific and therapeutic discoveries.

We introduce CRISPR-GPT, a solution that combines the strengths of LLMs with domain-specific knowledge, chain-of-thought reasoning, instruction fine-tuning, retrieval techniques and tools. CRISPR-GPT is centred around LLM-powered planning and execution agents (Fig. 1). This system leverages the reasoning abilities of general-purpose LLMs and multi-agent collaboration for task decomposition, constructing state machines and automated decision-making (Fig. 2a). It draws upon expert knowledge from leading practitioners and peer-reviewed published literature in gene editing for retrieval-augmented generation (RAG)[13].

Results

Building AI co-pilot harnessing LLM's reasoning ability

CRISPR-GPT supports four major gene-editing modalities and 22 gene-editing experiment tasks (Fig. 1 and Supplementary Table 1). It offers tunable levels of automation via three modes: Meta, Auto and Q&A. They are designed to accommodate users ranging from novice PhD-level scientists fresh to gene editing, to domain experts looking for more efficient, automated solutions for selected tasks (Fig. 1). The 'Meta mode' is designed for beginner researchers, guiding them through a sequence of essential tasks from selection of CRISPR systems, delivery methods, to designing gRNA, assessing off-target efficiency, generating experiment protocols and data analysis. Throughout this decision-making process, CRISPR-GPT interacts with users at every step, provides instructions and seeks clarifications when needed. The 'Auto mode' caters to advanced researchers and does not adhere to a predefined task order. Users submit a freestyle request, and the LLM Planner decomposes this into tasks, manages their interdependence, builds a customized workflow and executes them automatically. It fills in missing information on the basis of the initial inputs and explains its decisions and thought process, allowing users to monitor and adjust the process. The 'Q&A mode' supports users with on-demand scientific inquiries about gene editing.

To assess the AI agent's capabilities to perform gene-editing research, we compiled an evaluation test set, Gene-editing bench, from both public sources and human experts (details in Supplementary Note C). This test set covers a variety of gene-editing tasks (Fig. 1). By using the test set, we performed extensive evaluation of CRISPR-GPT's capabilities in major gene-editing research tasks, such as experiment planning, delivery selection, single guide (sg)RNA design and experiment troubleshooting. In addition, we invited human experts to perform a thorough user experience evaluation of CRISPR-GPT and collected valuable human feedback.

Further, we implement CRISPR-GPT in real-world wet labs. Using CRISPR-GPT as an AI co-pilot, we demonstrate a fully AI-guided knockout (KO) of four genes: *TGFβR1*, *SNAI1*, *BAX* and *BCL2L1*, using CRISPR-Cas12a in human lung adenocarcinoma cell line, as well as

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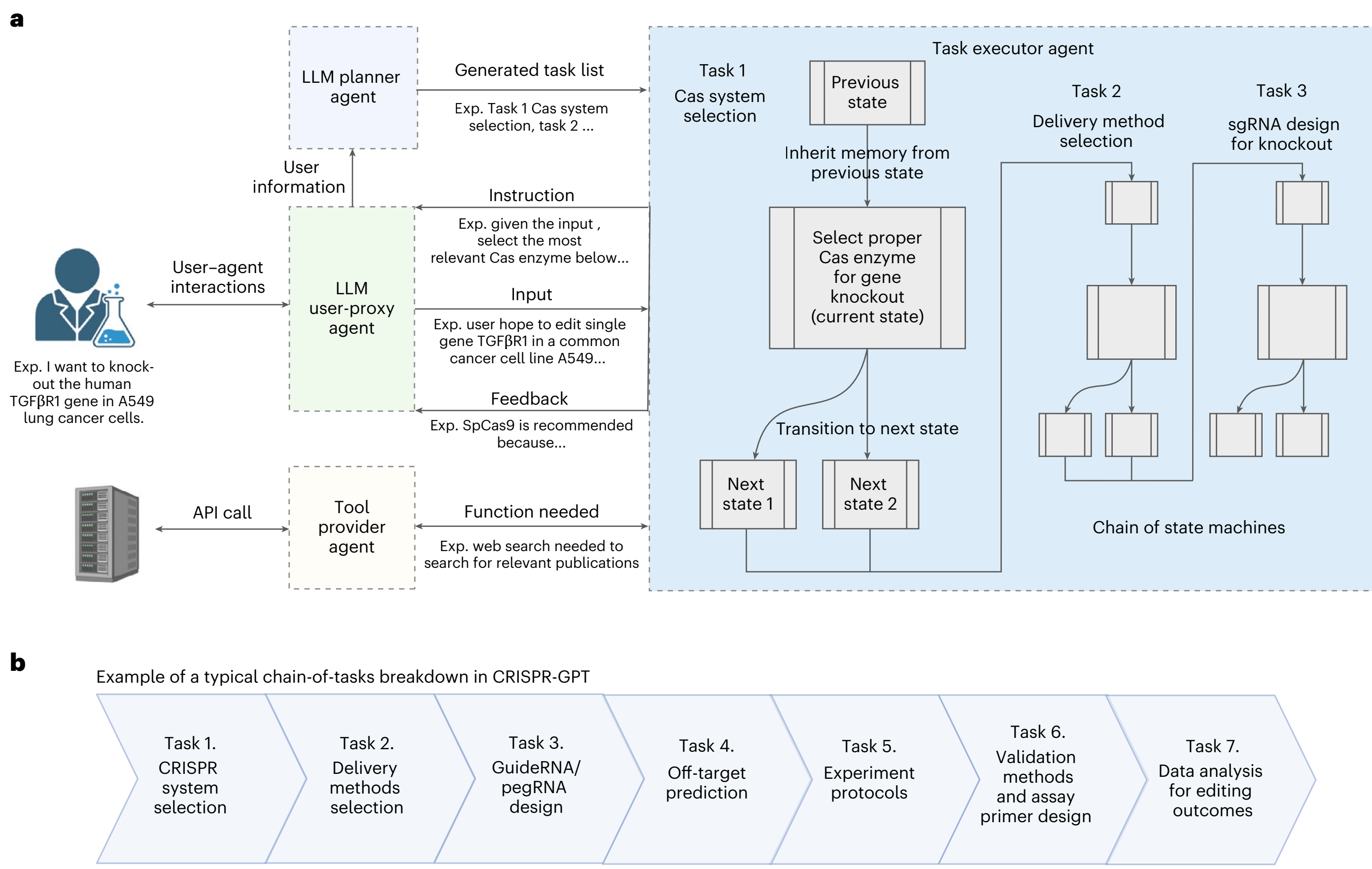

Fig. 2 | CRISPR-GPT adopts a compositional, multi-agent architecture to enable human–AI collaboration and automated experimental designs. a, The backbone of CRISPR-GPT involves multi-agent collaboration between four core components: (1) The LLM Planner agent is responsible for configuring tasks on the basis of the user's needs. It automatically performs task decomposition on the basis of the user's request, the descriptions of the currently supported tasks and internal knowledge. The state machines of the selected tasks are chained together to fulfill the user's request. (2) The Task executor agent implements the chain of state machines from the Planner agent and is responsible for providing instructions and feedback, receiving input from the User-proxy agent and calling external tools. State machines are central to the Task executor, where each state is responsible for one round of interaction with the user. The instruction is provided to the user first with sufficient information for the current decision-making step and the required inputs. After receiving the response from the user, it provides output and feedback, where Tool providers are potentially called during the execution of the state. Afterwards, the state machine transits to the next state. (3) The LLM User-proxy agent is responsible for interacting with the Task executor on behalf of the user, where the user can monitor the process and provide corrections to the User-proxy agent if the generated content needs modification or improvement. It generates responses to every step of the state machine on behalf of the user. (4) Tool providers support diverse external tools and connect to search engines or databases via API calls. Part of the panel was created with BioRender.com/svkmgjk. **b**, Breakdown of individual tasks in a typical CRISPR-GPT workflow for gene-editing experiments.

AI-guided CRISPR-dCas9 epigenetic activation of two genes: *NCR3LG1* and *CEACAM1*, in a human melanoma model. All these wet-lab experiments were carried out by junior researchers not familiar with gene editing. They both succeeded on the first attempt, confirmed by not only editing efficiencies, but also biologically relevant phenotypes and protein-level validation, highlighting the potential of LLM-guided biological research.

CRISPR-GPT is a multi-agent, compositional system involving a team of LLM-based agents, including an LLM Planner agent, a User-proxy agent, Task executor agents and Tool provider agents (Fig. 2a). These components are powered by LLMs to interact with one another as well as the human user. We also refer to the full system as an 'agent' to encapsulate the overall functionalities.

To automate biological experiment design and analysis, we view the overall problem as sequential decision-making. This perspective frames the interaction between the user and the automated system as a series of decision-making steps, each essential for progressing towards the ultimate goal. Take the Auto mode for example. A user can initiate the process with a meta-request, for example, "I want to knock out the human TGFβR1 gene in A549 lung cancer cells". In response, the agent's LLM Planner will analyse the user's request, drawing on its extensive internal knowledge base via retrieval techniques. Leveraging the reasoning abilities of the base LLM, the Planner generates a chain-of-thought[37,38] reasoning path and chooses an optimal action from a set of plausible ones while following expert-written guidelines. Consequently, the Planner breaks down the user's request into a sequence of discrete tasks, for example, 'CRISPR-Cas system selection' and 'gRNA design for knockout', while managing interdependencies among these tasks. Each individual task is solved by an LLM-powered state machine, via the Task executor, entailing a sequence of states to progress towards the specific goal. After the meta-task decomposition, the Task executor will chain the state machines of the corresponding tasks together into a larger state machine and begin the execution process, systematically addressing each task in sequence to ensure that the experiment's objectives are met efficiently and effectively (Fig. 2a).

The User-proxy agent is responsible for guiding the user throughout the decision-making process via multiple rounds of textual interactions (typical user interactions required by each task detailed in Supplementary Table 2). At each decision point, the internal state machine presents a 'state variable' to the User-proxy agent, which includes the current task instructions, and specifies any necessary input from the user to proceed. The User-proxy agent then interprets this state given the user interactions and makes informed decisions as input to the Task executor on behalf of the user. Subsequently, the User-proxy agent receives feedback from the Task executor, including the task results and the reasoning process that led to those outcomes. Concurrently, the User-proxy agent continues to interact with the user and provides them with instructions, continuously integrating their feedback to ensure alignment with the user's objectives (detailed in Methods; Fig. 2a and Supplementary Fig. 1).

To enhance the LLM with domain knowledge, we enable the CRISPR agent to retrieve and synthesize information from published protocols, peer-reviewed research papers and expert-written guidelines, and to utilize external tools and conduct web searches via Tool provider agents (Fig. 2a).

For an end-to-end gene-editing workflow, CRISPR-GPT typically constructs a chain of tasks that includes selecting the appropriate CRISPR system, recommending delivery methods, designing gRNAs, predicting off-target effects, selecting experimental protocols, planning validation assays and performing data analysis (Fig. 2b). The system's modular architecture facilitates easy integration of additional functionalities and tools. CRISPR-GPT serves as a prototype LLM-powered AI co-pilot for scientific research, with potential applications extending beyond gene editing.

CRISPR-GPT agents automate gene-editing research tasks

CRISPR-GPT is able to automate gene-editing research via several key functionalities. For each functionality we discuss the agentic implementation and evaluation results.

Experiment planning. The Task planner agent is charged with directing the entire workflow and breaking down the user's meta-request into a task chain (Fig. 2b). While the Planner selects and follows a predefined workflow in the Meta mode, it is able to take in freestyle user requests and auto-build a customized workflow in the Auto mode. For example, a user may only need part of the pre-designed workflow including CRISPR-Cas system selection, delivery method selection, gRNA design and experimental protocol selection before the experiment. Then the Task planner agent extracts the right information from the user request and assembles a customized workflow to suit user needs (Fig. 3a). To evaluate CRISPR-GPT's ability to correctly layout gene-editing tasks and manage intertask dependence, we compiled a planning test set, as part of the Gene-editing bench, with user requests and golden answers curated by human experts. Using this test set, we evaluated CRISPR-GPT in comparison with prompted general LLMs, showing that CRISPR-GPT outperforms general LLMs in planning gene-editing tasks (Fig. 3b). The CRISPR-GPT agent driven by GPT-4o scored over 0.99 in accuracy, precision, recall and F1 score, and had <0.05 normalized Levenshtein distance between agent-generated plans and golden answers (Fig. 3b). For extensive description of the test set and evaluation, please see Supplementary Note C1.

Delivery method selection. We present and evaluate the delivery agent of CRISPR-GPT (Fig. 4a,b). Delivery is a critical step for all gene-editing experiments. CRISPR-GPT equips LLM with expert-tailored instructions and external tools to choose delivery methods. Specifically, the agent first tries to understand the biological system that the user is planning to edit. It extracts keywords for the target cell/tissues/organisms, performs Google search and summarizes the results. Then, given its own knowledge and search results, CRISPR-GPT matches the user case with a major biological category (cell lines, primary cells, in vivo and so on) which reduces the possible options to a focused set of candidate methods. Next, CRISPR-GPT performs literature search with user and method-specific keywords, and ranks the candidate methods on the basis of citation metrics to suggest a primary and a secondary delivery method (Fig. 4a). To evaluate the performance of this module, we compiled test cases including 50 biological systems as part of the Gene-editing bench. For each case, we invited three human experts to score potential delivery options and utilized those as ground truth. We then evaluated the output of CRISPR-GPT and baseline models by comparing to the pre-compiled ground-truth score sheet. We found that CRISPR-GPT outperforms the baseline GPT-4 and GPT-3-turbo models (Fig. 4b). The agent has a substantial edge on difficult tasks such as those involving hard-to-transfect cell lines and primary cell types. We also noticed that including an additional literature search step improves the agent's performance only moderately. More details about the delivery selection evaluation can be found in Supplementary Note C2.

gRNA design. Good gRNA design is crucial for the success of CRISPR experiments. Various gRNA design tools and software, such as CRISPick[39–42] and ChopChop[43], are available. However, we believe there are two key challenges in general usage: (1) choosing a trustworthy source and (2) difficulty in quickly identifying gRNAs that suit specific user requirements or experiment contexts, often requiring lengthy sorting, ranking or literature review. To address these issues, we utilized pre-designed gRNA tables from CRISPick, a reputable and widely used tool. We leverage the reasoning capabilities of LLMs to accurately identify regions of interest and quickly extract relevant gRNAs. This approach is similar to the recently proposed 'chain-of-tables' methodology[44] (Fig. 4c, Extended Data Fig. 1a, and Supplementary Demo Videos 1 and 2). To evaluate the ability of CRISPR-GPT to correctly retrieve gRNAs, we compiled a gRNA design test set with ground truth from human experts (detailed in Supplementary Note C3). CRISPR-GPT agent outperforms the baseline LLMs in accurately selecting gRNA design actions and configuring the arguments (Fig. 4d).

Further, we picked a real-world test case from a cancer biology study, in which many highly ranked gRNA designs did not generate biological phenotypes, even when their editing efficiencies were high[45]. Instead, the authors of the study had to design gRNAs manually against exons encoding important functional domains within a gene, and exon-selected gRNAs induced expected cancer-killing effects. We tested CRISPR-GPT for designing gRNAs targeting BRD4 gene from this study, and compared results with those generated by CRISPick and CHOPCHOP (Extended Data Fig. 1). CRISPR-GPT was uniquely able to select the key exons, Exon3–4, within BRD4. In contrast, gRNAs designed by CRISPick or CHOPCHOP would be likely ineffective, as 7 out of 8 gRNAs mapped to non-essential exons (Extended Data Fig. 1). Taken together, our results support the benefit and validity of this module.

Other functions and tools. CRISPR-GPT provides specific suggestions on the choice of the CRISPR system, experimental and validation protocol by leveraging LLM's reasoning ability and retrieving information from an expert-reviewed knowledge base[46–51]. It also offers automated gRNA off-target prediction, primer design for validation experiments, and data analysis. In particular, the agent provides fully automated solutions to run external software, such as Primer3 (ref. 52), CRISPRitz[53–56] and CRISPResso2 (ref. 57) (Supplementary Table 1). We focused on implementing these tools as they are considered gold standard in respective tasks and have been extensively validated in previous work.

Problem solving via fine-tuning LLM on scientific discussion

General-purpose LLMs may possess broad knowledge but often lack the deep understanding of science needed to solve research problems. To enhance the CRISPR-GPT agent's capacity in answering advanced research questions, we build a Q&A mode that synthesizes information from multiple resources, including published literature, established protocols and discussions between human scientists, utilizing

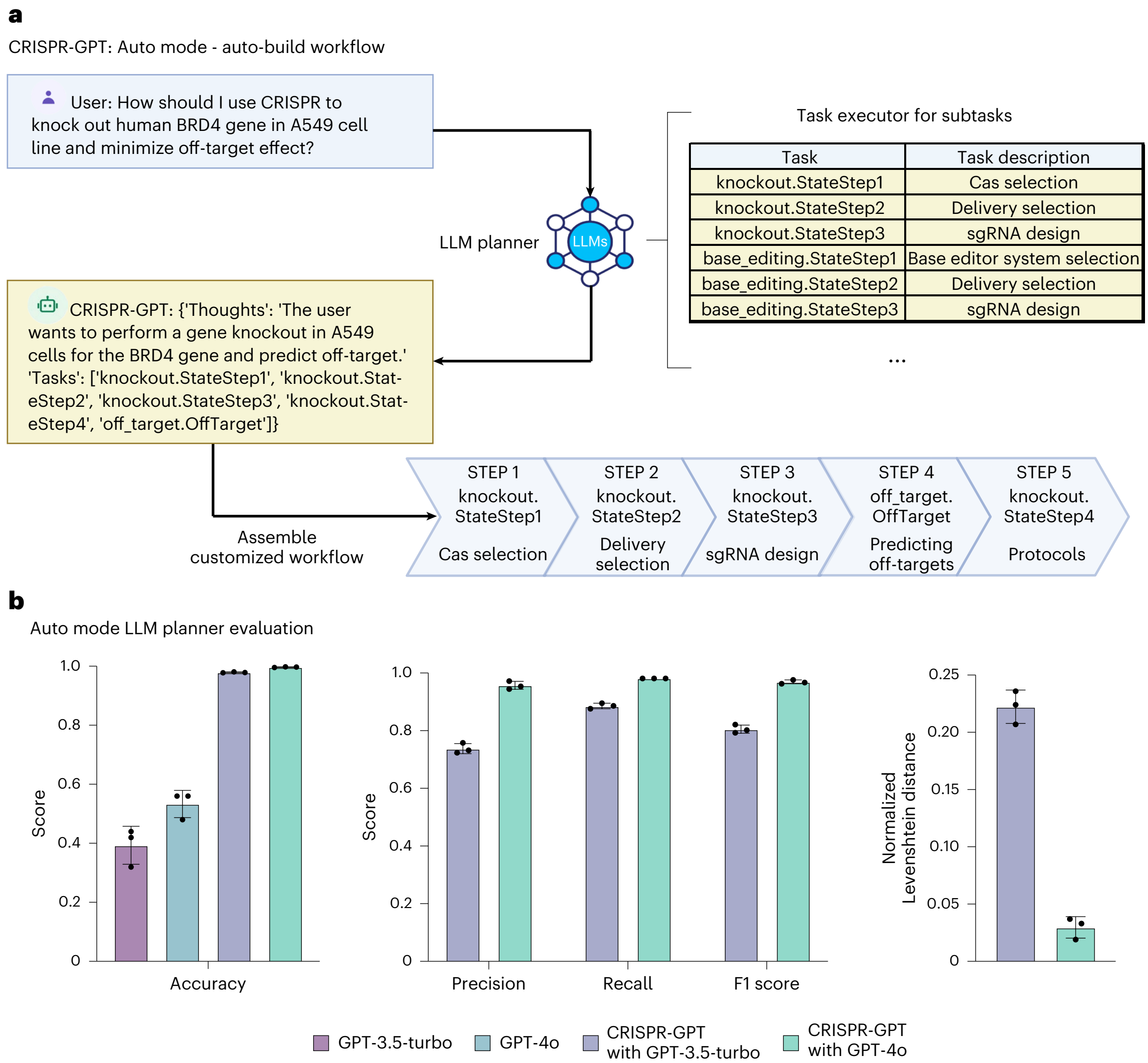

Fig. 3 | Task decomposition and experiment planning in CRISPR-GPT Auto mode with performance evaluation. a, The LLM Planner agent automatically breaks down the user's meta-request to a sequence of tasks. Then it assembles a customized workflow of the chained tasks to meet the user's needs. Part of the panel was created with BioRender.com/qy4v. **b**, Evaluation of the LLM Planner using a gene-editing planning test set. For each test case, we generate three independent answers from each model and report the average scores (Supplementary Note C). Data shown are the mean ± s.d. ($n = 3$ per group).

a combination of RAG technique, a fine-tuned specialized model and a general LLM (for which we picked GPT-4o; Methods).

To enhance the Q&A mode's capacity to 'think' like a scientist for problem solving, we sought to train a specialized language model using real scientific discussions among domain experts. The fine-tuned model is used as one of the multiple sources of knowledge for the Q&A mode (Fig. 4e). To this end, we collected 11 years of open-forum discussions from a public Google Discussion Group on CRISPR gene-editing, starting from 2013 (Supplementary Note B). The discussion group involved a diverse cohort of scientists worldwide. This dataset, comprising ~4,000 discussion threads, was curated into an instructional dataset with over 3,000 question-and-answer pairs (Supplementary Note B). Using this dataset, we fine tuned an 8-billion-parameter LLM on the basis of the Llama3-instruct model[58–65]. The fine-tuned model, which we call CRISPR-Llama3, has improved abilities in gene-editing questions, outperforms the baseline model on basic questions by a moderate 8% and on real-world research questions by ~20% (Supplementary Figs. 2 and 3). We integrate this fine-tuned LLM into the Q&A mode as a 'brainstorming source', enabling the agent to generate ideas like a human scientist and provide a second opinion for difficult queries (Fig. 4e).

To assess the performance of the Q&A mode, we used the Gene-editing bench Q&A test set (Supplementary Note C). The test questions encompass basic gene-editing knowledge, experimental troubleshooting, CRISPR application in various biological systems, ethics and safety[66,67]. We prompted CRISPR-GPT, GPT-3.5-turbo and GPT-4o to generate responses to test questions. Three human experts scored the answers in a fully blinded setting. The test demonstrated that the Q&A mode outperformed baseline LLMs in accuracy, reasoning and conciseness, with improvement of 12%, 15% and 32%, respectively, versus GPT-4o (Fig. 4f). Human evaluators observed that general-purpose LLMs sometimes make factual errors and tend to provide extensive answers not all of which are relevant to the questions. For example, one question was about solving cell growth issues in an experiment where a scientist performed Cas9 editing followed by single-cell sorting using MCF-7 cells. For this question, the Q&A mode provided a concise, accurate summary of potential reasons and actionable solutions. In contrast, GPT-4o responded with a long list of 9 factors/options, but at least 2 of them were not applicable to MCF-7 cells (Extended Data Fig. 2). This and other examples (Extended Data Figs. 3 and 4) showcase the advantage of the CRISPR-GPT Q&A mode. Overall, evaluation results confirmed that the multisource

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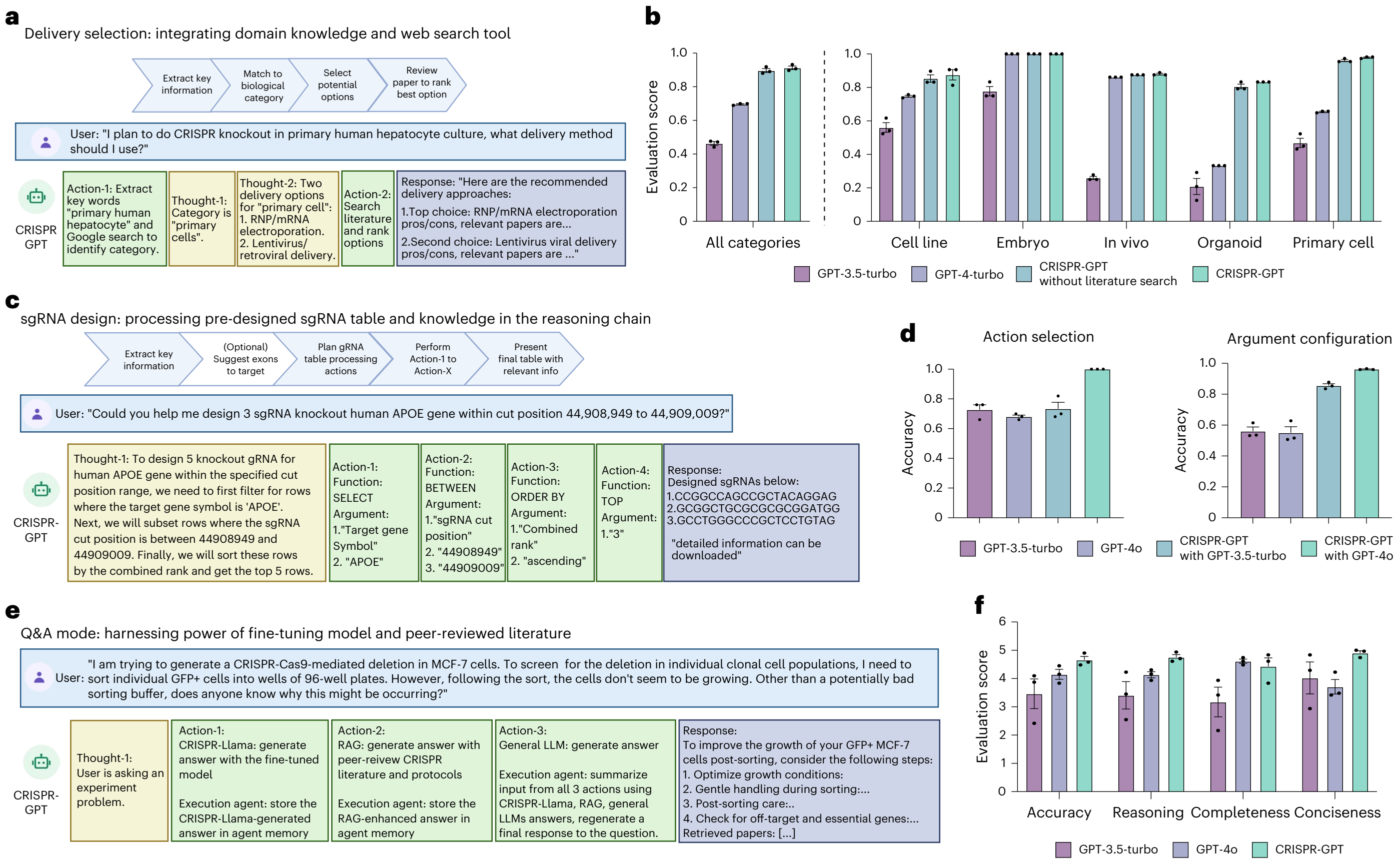

Fig. 4 | CRISPR-GPT automates gene-editing research and experiment tasks. **a**, Design of delivery method selection agent in CRISPR-GPT, showing the workflow, example request and a series of agent thoughts–actions to identify most suitable delivery methods for the user's needs. **b**, Evaluation results of delivery method selection using CRISPR-GPT and baseline models. Data shown are the mean ± s.d. ($n = 3$ per group). **c**, Design of gRNA design agent in CRISPR-GPT, showing the workflow, example request and a series of agent thoughts–actions to select top-ranked gRNA customized to user's request. **d**, Evaluation results of gRNA design using CRISPR-GPT and baseline models. Models were prompted to generate functions and associated parameters to design gRNAs requested by the user. Data shown are the mean ± s.d. ($n = 3$ per group). **e**, Design of Q&A mode in CRISPR-GPT, showing the workflow, example request and a series of agent thoughts–actions to answer gene-editing questions. **f**, Evaluation of CRISPR-GPT and baseline models for answering gene-editing research questions. Models were prompted to generate answers, which were anonymized, evaluated by three human experts in a fully blind setup. Scores range from 1 (lowest) to 5 (highest). All scores above were from three independent trials (details on evaluation in Supplementary Note C). Data shown are the mean ± s.d. ($n = 3$ per group).

Q&A mode is better at answering advanced research questions about gene editing.

CRISPR-GPT excels in human–AI collaboration

To further evaluate the human user experience of CRISPR-GPT, we assembled a panel of eight gene-editing experts to assess the agents' performance for end-to-end experiment covering all 22 individual tasks (Supplementary Table 2 and 3 demos). The experts were asked to rate their experiences in four dimensions: accuracy, reasoning and action, completeness and conciseness (see Supplementary Note C for detailed rubrics). CRISPR-GPT demonstrated improved accuracy and strong capabilities in reasoning and action, whereas general LLMs, such as GPT-4o, often included errors and were prone to hallucination (Fig. 5a,b).

Highlighted by human evaluators' observations (Fig. 5c), the CRISPR-GPT agent provides users with more accurate, concise and well-rounded instructions to execute the planned experiments. The ability of CRISPR-GPT to perform specialty gene-editing tasks, such as exon-selected gRNA design, customized off-target prediction and automated sequencing data analysis, reinforced its advantage versus general-purpose LLMs. This is confirmed by the task-specific evaluation results (Fig. 5b). Despite its strengths, CRISPR-GPT struggled with complex requests and rare biological cases, highlighting areas for improvement (limitations in Supplementary Note D).

Real-world demonstration for fully AI-guided gene editing

To showcase and validate CRISPR-GPT's ability as an AI co-pilot to general biologists, we enlisted two junior researchers unfamiliar with gene editing. They used CRISPR-GPT in two real-world experiments: (1) designing and conducting a multigene knockout and (2) epigenetic editing, from scratch.

In the first experiment, the junior researcher conducted gene knockouts in the human A549 lung adenocarcinoma cell line, targeting four genes involved in tumour survival and metastasis: *TGFβR1*, *SNAI1*, *BAX* and *BCL2L1* (Fig. 6). The experiment was designed from scratch with CRISPR-GPT (Fig. 6a). On the basis of the user–AI interaction, enAsCas12a was selected for its multitarget editing capability and low off-target effects. For delivery, CRISPR-GPT recommended lentiviral transduction for stable Cas and gRNA expression. The gRNAs for the four target genes were designed through CRISPR-GPT. Furthermore, CRISPR-GPT provided step-by-step protocols for gRNA cloning, lentivirus production and viral delivery into A549 cells. To validate the editing, the researcher followed CRISPR-GPT's next-generation sequencing (NGS) protocol, using assay primers designed via the integrated Primer3 tool. After generating the NGS data, the raw sequencing files were uploaded into CRISPR-GPT for automated analysis through the CRISPResso2 pipeline. The analysis reports, sent directly via email, summarized the editing outcomes and showed consistently ~80% high efficiency across all target genes (Fig. 6b, Supplementary Demo

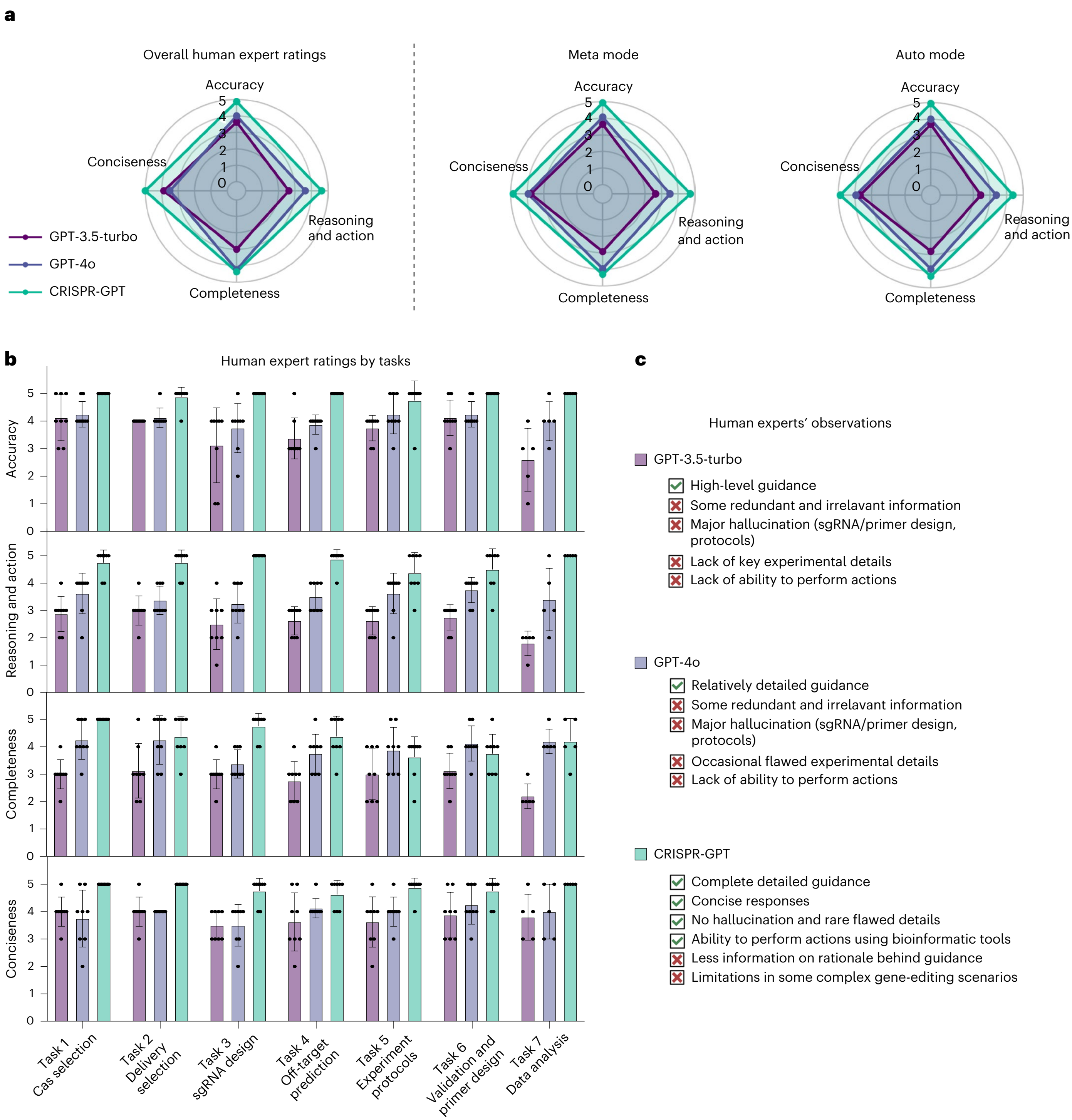

Fig. 5 | CRISPR-GPT outperforms general-purpose LLM for gene-editing research in human user experiences. a, Human user experience: evaluation of CRISPR-GPT for end-to-end gene-editing copiloting. Human experts scored performances from 1 (lowest) to 5 (highest). See detailed procedure and rubrics in Supplementary Note C (full chat history and video demo provided in Supplementary Table 2). **b**, Human user experience: evaluation results breakdown by major gene-editing tasks. Data shown are the mean ± s.d. ($n = 8$ per group). **c**, User observations on the strengths and limitations of CRISPR-GPT compared to baseline LLMs.

Video 3, user interactions summarized in Supplementary Table 4, full chat history listed in Supplementary Table 2). To further assess the biological phenotypes of *TGFβR1* and *SNAI1* knockout in A549 cells, the researcher conducted an epithelial–mesenchymal transition (EMT) induction experiment by treating A549 cells with TGFβ (Fig. 6c and Methods). The qPCR results revealed that the knockout A549 cell lines (A549 TGFβR1 KO and A549 SNAI1 KO) showed up to 9-fold reduction in CDH1 expression change and up to 34-fold reduction in VIM expression change, which are both key marker genes in the EMT process. This confirms the biological role of TGFβR1 and SNAI1 signalling in driving EMT progression (a crucial driver of metastasis) in lung cancer cells (Fig. 6d).

In the second experiment, the junior researcher performed epigenetic editing to activate two genes involved in cancer immunotherapy resistance in a human melanoma model cell line (Fig. 6e, user

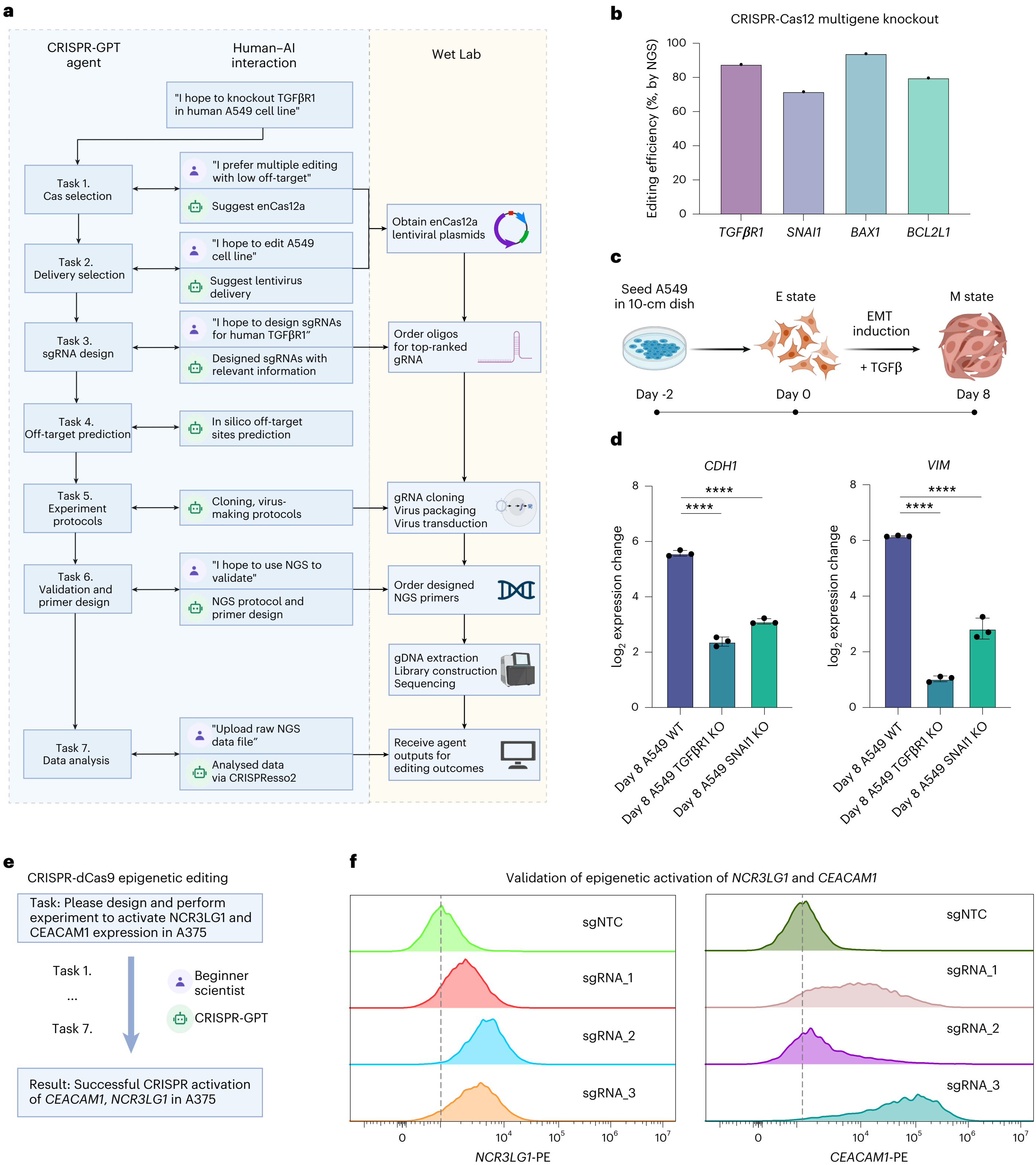

Fig. 6 | Wet-lab demonstrations of CRISPR-GPT in knockout and activation experiments. a, The full workflow of CRISPR-GPT-guided knockout experiment of *TGFβR1*, *SNAI1*, *BAX1* and *BCL2L1* through multiple rounds of human–AI interaction (*TGFβR1* knockout is shown as an example, see Supplementary Demo Video 3 and full chat history listed in Supplementary Table 2). Panel was originally created with BioRender.com/iim1m2t and processed. **b**, Editing efficiencies for *TGFβR1*, *SNAI1*, *BAX1* and *BCL2L1* measured via NGS, analysed using CRISPResso2 and CRISPR-GPT. **c**, Schematic of the EMT induction process via TGFβ treatment (Methods). Panel was originally created with BioRender.com/iim1m2t and processed. **d**, Functional outcomes of *TGFβR1* and *SNAI1* knockout in A549 cells after EMT induction by TGFβ. qPCR analysis shows reduced expression changes in EMT marker genes (*CDH1*, *VIM*) in A549 *TGFβR1* and *SNAI1* knockout cells compared with A549 wildtype (WT) cells, confirming successful knockouts of *TGFβR1*/*SNAI1*. Data shown are the mean ± s.d. ($n = 3$ per group). One-way analysis of variance (ANOVA) was used to calculate statistics. ****$p < 0.0001$. **e**, Simplified workflow of a beginner researcher activating *NCR3LG1* and *CEACAM1* expression through multiround interactions with CRISPR-GPT (full chat history listed in Supplementary Table 2). **f**, Editing outcomes of *NCR3LG1* and *CEACAM1* activation using CRISPR-GPT-designed sgRNAs, measured via flow cytometry (Methods).

interactions summarized in Supplementary Table 4, full chat history listed in Supplementary Table 2). CRISPR-GPT guides the researcher through the full workflow: identifying the most suitable CRISPR activation system, selecting an appropriate delivery method for A375 cells, designing dCas9 gRNAs (three gRNAs per gene) and generating protocols for validating editing outcomes. After editing was completed, measurements of target protein expression level confirmed successful activation of both genes, with up to 56.5% efficiency for *NCR3LG1*, and 90.2% efficiency for *CEACAM1*, when comparing gRNA-edited groups vs negative control gRNAs (Fig. 6f).

Overall, CRISPR-GPT enabled successful completion of the first set of AI-guided gene-editing experiments. Interactions between the researchers and LLM-powered agents led to efficient, accurate and ethically mindful gene-editing on the first attempt, even by users new to the technique.

Discussion

CRISPR-GPT demonstrates the potential of LLMs to automate and enhance biological research and experiment design and analysis. This AI-guided workflow leverages LLM for reasoning, multi-agent collaboration and scientific discussions for brainstorming, reduces errors, and improves research quality and reproducibility. Despite its current capabilities, CRISPR-GPT has limitations. For example, the agent system relies on high-quality instructions and discussion data from human scientists who have deep knowledge about the biology domain. Such data are hard to collect, creating challenges for further improvements and scaling up. Further, evaluation of such AI tools is generally challenging due to the need to collect substantial feedback from human biologists. For another example, the current gRNA design step mainly supporting human and mouse targets could be further expanded.

Technologies such as CRISPR-Cas9 pose potential ethical and safety risks, including potential misuse for dual purposes, which can be exemplified with AI tools[68]. Altering human genomes raises substantial ethical concerns, particularly with germline cell and embryo editing. Due to these concerns, such editing is illegal in the United States and many other countries. In addition, gene-editing technology could be abused to create bioweapons, such as genetically engineered viruses[69].

To mitigate these risks, we augment CRISPR-GPT with an additional layer of safety mechanism to defend against malicious dual uses. Following the guidelines given in a moratorium[70] on heritable genome editing, CRISPR-GPT ensures that users cannot bypass the step of specifying the organism they are editing. If the target is human tissue or organs, the system triggers the following steps: (1) displays a warning note when proceeding with human gene-editing experiments; (2) provides a link to the international moratorium with an explanatory note; and (3) asks users to confirm that they understand the risks and have read the international guidelines before proceeding. The agent also checks if the user request involves editing of human germline cells or dangerous, pathogenic viruses (Supplementary Note D and Data 2). If such a risk is identified, it will trigger an error message and stop proceeding (Extended Data Fig. 5 for examples of the risk mitigation tests).

Other concerns are related to user data privacy issues, especially when human genome sequence information might be exchanged by using AI tools. We follow the data privacy and HIPAA privacy rule in healthcare[71]. Although genome-scale sequences are fundamentally linked to identities, DNA segments of up to 20-bp length are considered safe and not able to identify human identity[72].

CRISPR-GPT includes functionalities to prevent sharing of identifiable private human/patient sequences with public LLM models. Our solution involves two key measures. First, the system would never store any identifiable long genome sequence in the server that would potentially reveal patient private information. Second, a filter is implemented to detect any sequence of 20 or more A/T/G/C/U bases in prompts before sending them to external LLMs. If detected, the agent raises an error with a warning note, asking the user to manually remove the sequence from the input. This prevents the leakage of sensitive information to external models and tools (Extended Data Fig. 5).

Looking ahead, the uses of CRISPR-GPT could be further expanded by connecting to latest advances in genome/protein foundation models, plasmid design tools and other machine learning models, to enable experiment design and analysis tasks beyond gene editing. In addition, the integration of CRISPR-GPT with automated laboratory platforms and robotics holds immense promise. By bridging computational design, analysis and physical execution, researchers could leverage the agent's expertise to orchestrate end-to-end automated experiments, minimizing manual intervention and accelerating the pace of discovery.

Methods

Large language model-powered autonomous agent

CRISPR-GPT consists of the following 4 core components (Fig. 1): LLM Planner, Tool providers, Task executors, and the LLM User-proxy agent that serve as the interface with users for taking inputs and communicating outputs. Each component can be viewed as an LLM-powered single agent with relatively simple functionality, and the overall system functions via multi-agent interaction. These single agents leverage general-purpose LLMs, such as GPT-4o (used in all four core agents unless otherwise specified, such as when comparing GPT-4 vs GPT-3.5, where we use GPT-4o and GPT-3.5), as their base model to handle a wide range of tasks. The LLMs rely on carefully designed prompts to guide their behaviour and interactions, with example prompts provided in Supplementary Note E.

The Task executor operates as state machines, providing robust decomposition and progress contro

A total of 22 tasks (summarized in Supplementary Table 1) have been implemented, each decomposed into subgoals, with states providing instructions and guiding users through decision-making via multiple rounds of textual interaction. A central management class tracks the current state, task queue, and memory for state outputs and execution history. State transitions occur sequentially, or based on conditional logic from execution results as needed. States process user input and generate structured outputs containing the status, reasoning and response, which are stored continuously to ensure continuity across tasks. This framework supports both predefined workflows (Meta mode) and dynamically generated task sequences (Auto mode), offering flexibility, reliability and robust error handling for executing CRISPR-GPT tasks.

In Meta mode, the Task executor follows predefined workflows that cover complete pipelines for four meta-tasks, each corresponding to a major type of gene-editing experiment. In Auto mode, the LLM Planner dynamically generates a customized sequence of tasks on the basis of the user's meta-request. The Task executor then constructs and executes the workflow by chaining the state machines of the corresponding tasks into a larger state machine, enabling seamless and automated execution of complex gene-editing pipelines.

The Tool provider connects the Task executor with external application programming interfaces (APIs)

To connect language models with external functionalities[73–77], the system needs to (1) analyse the current situation and judge whether it is suitable to call an external tool; and (2) know what kinds of tools are available and choose the best from them. Instead of directly exposing the interfaces of the APIs to LLMs, in CRISPR-GPT, we wrap the usage of APIs inside the states and expose more user-friendly and LLM-friendly textual interfaces through hand-written instructions and responses. In plain words, we are teaching users (human agents and LLM User-proxy agents) to use the tools. The tools include Google web search, Google Scholar search, literature retrieval and bioinformatic tools such as Primer3 (ref. 52), CRISPRitz[53] and CRISPResso2 (ref. 57).

The LLM Planner automatically plans gene-editing experiments on the basis of the user's request

LLMs such as GPT-4, Gemini and Claude serve as the reasoning core of the LLM-powered agent to solve real-world decision-making problems. Our LLM Planner operates on the basis of two key components: (1) the user query and (2) a predefined table containing comprehensive descriptions and interdependency information for all available tasks (example in Supplementary Note E). Using the ReAct prompting technique, the LLM is prompted to output a chain-of-thought reasoning path along with the final action from the plausible action set (Fig. 2a). On the basis of LLM's internal knowledge, combined with our manually written task descriptions and decomposition instructions, the Planner analyses the user's request, intelligently decomposes it into an ordered list of tasks, and ensures that the dependencies between tasks are respected (detailed prompt format provided in Supplementary Note E). Once the decomposition is complete, the corresponding state machines are automatically chained together to execute all tasks in the appropriate sequence. For robustness, we prevent the LLM from dynamically adding or deleting tasks (state machines) during execution. However, we acknowledge that enabling dynamic task management is an important step towards developing a more intelligent science AI agent and leave this for future work.

LLM User-proxy agent automatically interacts with the Task executor on the basis of the meta-request

The LLM User-proxy agent automatically interacts with the Task executor on the basis of the meta-request. Central to our system, this agent serves as an intermediary between the user and a state machine derived from an initial task decomposition step—breaking down the gene-editing process into a structured sequence of actions and decisions. At each step, the state machine presents a current state to the LLM agent, which includes a task description and specifies any required input. This input varies by task type and may include general experimental context (for example, "I hope to design 4 sgRNAs targeting human TGFβR1") or a specific Cas system (for example, enCas12a), as shown in Supplementary Fig. 1b.

The LLM User-proxy agent interprets the current state and makes informed decisions on the user's behalf. It integrates multiple sources of information, including:

1. Instructions from the state machine,
2. User requests,
3. Session interaction history, and
4. Results from external computational tools.

This synthesized information is formatted into a structured prompt for the agent, which determines the most appropriate next action. For instance, when designing a CRISPR experiment, the agent might combine user input about a target gene with computational results to suggest sgRNA candidates.

While the User-proxy agent operates autonomously, user oversight remains essential. Users are encouraged to monitor task progression and intervene as needed to correct errors or misinterpretations, preserving the integrity of the gene-editing design (Supplementary Fig. 1a).

This approach fosters a collaborative synergy between human expertise and AI. By leveraging the LLM agent's reasoning ability, we enable a more efficient, accurate and user-friendly design experience. The sequential decision-making framework streamlines execution while ensuring that user input remains central to experiment planning.

Delivery method selection agent

Our approach mirrors the thought process of human gene-editing experts to identify the most appropriate delivery method on the basis of the user's specific biological system. The workflow is illustrated in Fig. 4a. It begins by instructing the LLM to extract key biological terms from the user's natural language request. These terms provide insight into the biological context of the experiment. The LLM is then tasked with accessing up-to-date information using a Google Search API to gather additional context about the biological system in the user request.

On the basis of the combined information from the user's request and external data, the LLM categorizes the system into one of 7 major biological categories:

1. Mammalian in vivo
2. Mammalian embryos
3. Mammalian primary cells or stem cells ex vivo
4. Mammalian cell lines with strong evidence of high-efficiency transfection
5. Mammalian cell lines or organoids without strong evidence of high-efficiency transfection
6. Human in vivo or human embryos
7. Bacteria, viruses and other organisms

These categories encompass the majority of biological systems relevant to CRISPR delivery. For each category, we curated 1–3 delivery methods on the basis of human experts' knowledge, which represent the most commonly used CRISPR delivery strategies.

To further tailor the recommendations to the user's specific scenario, the agent system conducts a Google Scholar search to identify relevant peer-reviewed literature. The search is guided by the key terms extracted from the user's request. From the search results, the top 10 relevant papers are ranked by citation count, providing a quantitative measure for prioritizing the potential delivery options within each biological category.

While citation numbers are not a definitive metric for determining the most appropriate delivery method, they offer a useful reference point. This approach helps to present well-informed recommendations along with relevant literature to the user.

Guide RNA design agent

Designing sgRNAs is a critical challenge in CRISPR editing, as it directly impacts editing efficiency. While many sgRNA design tools (web-based and software packages) exist, they typically follow shared design principles and use metrics, such as on-target/off-target scores, exon number and cut position, to rank candidates. We identified two main user challenges: (1) finding a trustworthy sgRNA design source and (2) efficiently selecting sgRNAs that meet specific criteria without manually evaluating every metric.

To address these issues, we leveraged pre-designed sgRNA tables from CRISPick, a widely used and validated library from the Broad Institute. We combined this with the reasoning and action (ReAct) capabilities of LLMs to process user-driven table queries. Our agent executes a series of actions step by step to generate outputs, akin to the recently described 'chain-of-table' methodology[15].

The agent system can choose from four key functions:

- SELECT: Retrieves rows where the specified column matches the given value.
- BETWEEN: Selects rows where the specified column's values fall within a specified range (inclusive).
- ORDERBY: Orders the table on the basis of values in a specified column.
- TOP: Returns the top *N* rows of the table.

These can be expanded in the future via human or LLM-driven suggestions. The agent extracts relevant parameters from both the user request and table, applies the functions, and returns pre-designed sgRNAs with associated metadata through a visual table and download link.

We also developed an optional exon suggestion module for CRISPR knockout design. Since sgRNAs targeting non-essential regions can be less effective, we hypothesized that LLMs could use their broad knowledge base to suggest functionally important exons. For example, targeting the BD1/BD2 domains has been shown to effectively disrupt BRD4 function[44]. We prompt the LLM to reason through gene function and recommend candidate exons (Extended Data Fig. 1), which are then incorporated into table queries.

To our knowledge, no existing sgRNA design tools integrate gene functional domains, making this exon suggestion feature a valuable reference. However, we note that its performance may be limited for genes with sparse literature or minimal online information.

Q&A mode

General-purpose LLMs often lack sufficient understanding of advanced biology. As outlined in Supplementary Note A, we identified key failure modes: (1) information hallucination, (2) outdated CRISPR knowledge, (3) absence of peer-reviewed sources and (4) poor alignment with user-specific problem-solving needs. To overcome these challenges, CRISPR-GPT's Q&A mode employs a multisource system for answering advanced biology questions (Fig. 4e). Upon receiving a query, it synthesizes information from three sources:

1. Fine-tuned CRISPR-LLama: Trained on human discussion threads from a CRISPR-focused Google Group, this model improves problem-solving and troubleshooting beyond the baseline (Supplementary Note B).
2. RAG-based literature retrieval (Tool provider agent): This accesses a curated, expert-selected CRISPR literature database (~50 key papers chosen for impact and recency; Supplementary Fig. 4 and Note F). Using OpenAI Embeddings and FAISS, both database entries and user queries are embedded into semantic vectors. The top k ($k = 4$) passages are retrieved by cosine similarity, ranked and summarized to guide the model's response.
3. General-purpose LLM: For example, ChatGPT or LLama, used as an additional source.

Extendibility of CRISPR-GPT

Given that CRISPR-GPT has a modular multi-agent architecture, integrating new tools and functions into the existing system is easy and training free. To add a new tool/function, the procedure is as follows:

(1) Tool wrapping: Develop specific code to encapsulate the tool's functionality within a state machine, which we call a Tool provider agent. This wrapper presents user-friendly and LLM-friendly textual interfaces through carefully crafted instructions and responses.
(2) Meta mode integration: If we want to add the tool to be used in the Meta mode, we add the entry state of the new state machine to appropriate positions within the relevant predefined workflow.
(3) Auto mode integration: Register the entry state of the new tool's state machine in the task decomposition table. This ensures that during task decomposition, the Planner agent becomes aware of the new tool and can incorporate it into its decision-making process.

Performance assessment of CRISPR-GPT

Benchmark dataset. We compiled Gene-editing bench, a collection of test questions and answers for evaluating AI tools' capabilities for CRISPR experimental design, with a total of 288 unique entries covering four topics:

1. Gene-editing planning: we compiled a total of 50 test cases and answers curated by consensus of human gene-editing experts.
2. CRISPR gRNA design: 50 test cases with pre-compiled answers by human experts.
3. Gene-editing delivery method selection: 50 test cases covering a range of biological systems and major experiment types. For each test case, we asked human experts to rank the available delivery method and report the consensus ranking as answer.
4. Gene-editing Q&A: 138 questions and answers, filtered for errors or issues, compiled from both public sources and human experts.

Validation of individual gene-editing agents

Using this benchmark dataset, we evaluated key functions of the CRISPR-GPT agent system:

1. Planning evaluation: For each query, we generated three batches of subtask lists using CRISPR-GPT and compared them to ground truth using accuracy, precision, recall and F1 scores. Task ordering was assessed via normalized Levenshtein distance. We also tested GPT-4o and GPT-3.5-turbo for comparison, evaluating the models' ability to plan and sequence gene-editing tasks.
2. Delivery method selection: For each test case, CRISPR-GPT (with and without literature search), GPT-3.5-turbo and GPT-4-turbo proposed primary and secondary delivery methods. Responses were scored against ground truth (primary: weight 2; secondary: weight 1), summed across categories to assess each model's ability to suggest delivery methods across biological systems.
3. Guide RNA design evaluation: CRISPR-GPT generated gRNA design function lists and parameters, which were compared to ground truth to evaluate function selection, order and parameter accuracy. We also tested GPT-4 and GPT-3.5-turbo, assessing their ability to interpret user intent and produce effective design strategies.
4. Q&A mode evaluation: We selected 40 questions and prompted CRISPR-GPT, GPT-3.5-turbo and GPT-4 to generate answers. Three human experts scored responses across four aspects in a blinded setup, and average scores were used to determine final performance, evaluating the models' ability to handle diverse gene-editing questions.

Detailed evaluation procedures for all the above are provided in Supplementary Note C.

Human experience evaluation

To holistically evaluate user experiences of the CRISPR-GPT, we invited 8 independent CRISPR human experts to test the agent system via its web surface. Each expert was asked to make one gene-editing request under the Meta mode and two gene-editing requests under the Auto mode. More details on the evaluation procedures are given in Supplementary Note C. In addition, we also provide a total of 20 full chat history demos from these tests in Supplementary Data 1 (details listed in Supplementary Table 2).

Real-world applicability of CRISPR-GPT via wet lab demonstrations

To evaluate the real-world applicability of CRISPR-GPT, we conducted two independent wet lab demonstrations:

1. Beginner Researcher 1: We invited an independent junior PhD scientist unfamiliar with the CRISPR field to perform CRISPR gene-editing experiments using CRISPR-GPT via human–agent collaboration. The researcher applied CRISPR-GPT to execute a gene knockout experiment as part of a cancer research project. The agent provided step-by-step guidance throughout the process (video demonstrations are available

in Supplementary Demo Videos 1–3, full chat history in Supplementary Data 1, and details in Supplementary Table 2). The results were validated through next-generation sequencing and functional assays.
2. Beginner Researcher 2: An undergraduate student, also unfamiliar with the CRISPR field, was invited to perform gene-editing experiments through collaboration with CRISPR-GPT. The student implemented CRISPR activation in a cancer immunology research project, with stepwise guidance provided by the agent (full chat history provided in Supplementary Data 1, details in Supplementary Table 2). The results were validated through antibody staining and FACS sorting.

Cell line and cell culture
A375 and A549 cells were cultured in DMEM medium (high glucose, GlutaMAX; 10-569-044 Gibco) with 10% fetal bovine serum (FBS; 100-106, Gemini Bio), 100 U ml^{-1} penicillin and 100 μg ml^{-1} streptomycin (15140-122 Gibco). Cells were maintained at 37 °C in a humidified atmosphere with 5% CO_2.

CRISPR RNA cloning
Cloning of sgRNAs was carried out using BbsI or Esp3I (R3539, R0734 NEB) through a Golden Gate assembly into a lentiviral backbone. The constructs were sequence-verified via Sanger sequencing using a U6 sequencing primer (5′-GACTATCATATGCTTACCGT-3′).

Lentivirus packaging and transduction
Lentivirus production was performed by co-transfecting the assembled lentiviral vector with VSV-G envelope and Delta-Vpr packaging plasmids into HEK-293T cells using PEI transfection reagent (765090, Sigma-Aldrich). Supernatants were collected at 48 h post transfection. A375 and A549 cells were transduced at low multiplicity of infection (MOI) with 8 μg ml^{-1} polybrene using a spin infection method at 1,000 × *g* for 45 min. After 24 h, cells were selected with 1 μg ml^{-1} puromycin to establish stable cell lines.

Genomic DNA extraction, PCR and sequencing
Genomic DNA was extracted from selected cells at 7 days post transfection using QuickExtract DNA Extraction Solution (LGCQE09050, Lucigen) following manufacturer instructions. Targeted loci were amplified via PCR using Phusion Flash High-Fidelity PCR Master Mix (F548L, ThermoFisher) with primers containing Illumina sequencing adapters. Paired-end reads (150 bp) were generated using the Illumina MiSeq platform.

PCR primers:

- TGFβR1: Forward: AGATAGAGGGTACTACGTTGAAAGACT; reverse: AAAAAAGTCTTTCAACGTAGTACCCTCT
- SNAI1: Forward: AGATCAGTTGAAGGCCTTTCGAGCCTG; reverse: AAAACAGGCTCGAAAGGCCTTCAACTG
- BAX: Forward: AGATATCCAGGATCGAGCAGGGCGAAT; reverse: AAAAAATTCGCCCTGCTCGATCCTGGAT
- BCL2L1: Forward: AGATACGCACAGTGCCCCGCCGAAGGA; reverse: AAAATCCTTCGGCGGGGCACTGTGCGT

TGFβ treatment
For optimal EMT induction, cells were seeded at a density of 750,000 cells per 100 mm tissue culture plate and incubated for 24 h. The medium was then replaced with 2% FBS DMEM for an additional 24 h. Cells were subsequently treated with 5 ng ml^{-1} TGFβ (240-B/CF, R&D) in 2% FBS DMEM for 7 days. To ensure consistent cell density during the treatment, cells were reseeded at the same density every 2 days.

Quantitative PCR
Total RNA was extracted using the Direct-zol RNA Purification kit (R2051, Zymo Research) according to manufacturer instructions. Complementary DNA synthesis and qPCR were performed using the Power SYBR Green RNA-to-CT 1-Step kit (4389986, ThermoFisher) on a BioRad CFX384 system (BioRad). Gene expression was quantified using specific primers for *CDH1* (forward: CTG AGG ATG GTG TAA GCG ATG; reverse: GTC TGT CAT GGA AGG TGC TC) and *VIM* (forward: GTG AAT CCA GAT TAG TTT CCC TCA; reverse: CAA GAC CTG CTC AAT GTT AAG ATG). Expression levels were normalized to appropriate housekeeping genes.

Flow cytometry (FACS) analysis
To assess the expression of NCR3LG1 and CEACAM1, cells were stained with B7-H6 monoclonal antibody (JAM1EW), PE (eBioscience) (1:100 dilution) for NCR3LG1 and anti-CD66a/c/e mouse monoclonal antibody (PE [Phycoerythrin], clone ASL-32, 1:100 dilution) for CEACAM1. Staining was performed following manufacturer guidelines, and data were acquired using a CytoFLEX analyser. Flow cytometry data were analysed using standard software. Detailed gating strategies are shown in Supplementary Fig. 5.

Reporting summary
Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability
The main data supporting the results in this study are available within the paper and its Supplementary Information. Additional data are deposited in GitHub at https://github.com/cong-lab/crispr-gpt-pub (ref. 78).

Code availability
Because of safety concerns, data, code and prompts will not be fully released to the public until the development of US regulations in the field of artificial intelligence and its scientific applications. While the full code is not freely available, it has been peer reviewed. We will also release a light version in GitHub at https://github.com/cong-lab/crispr-gpt-pub (ref. 78).

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Author contributions

Y.Q., K.H., M.Y., M.W. and L.C. designed the research, constructed models, performed experiments and analysed data. K.Z., D.L. and D.Z. supported model construction, deployment and evaluation. Y.Q., X.W., D.Y, M.Y., H.C.C., W.A.J., M.S. and R.B.A. contributed to human evaluation and analysed data. L.C. and M.W. conceived and supervised the research. Y.Q., K.H., M.Y., M.W. and L.C. wrote the paper with input from all authors.

Competing interests

Princeton University and Stanford University have filed patent applications (#19/093,928, Princeton and Stanford, 2025) based on this work, where L.C., M.W., Y.Q. and K.H. are listed as inventors. L.C. is a member of the scientific advisory board of Arbor Biotechnologies. L.C. has equity interest in Auto Bio, Rootpath Genomics, and Acrobat Genomics. D.Z. is an employee of Google DeepMind. The remaining authors declare no competing interests.

Additional information

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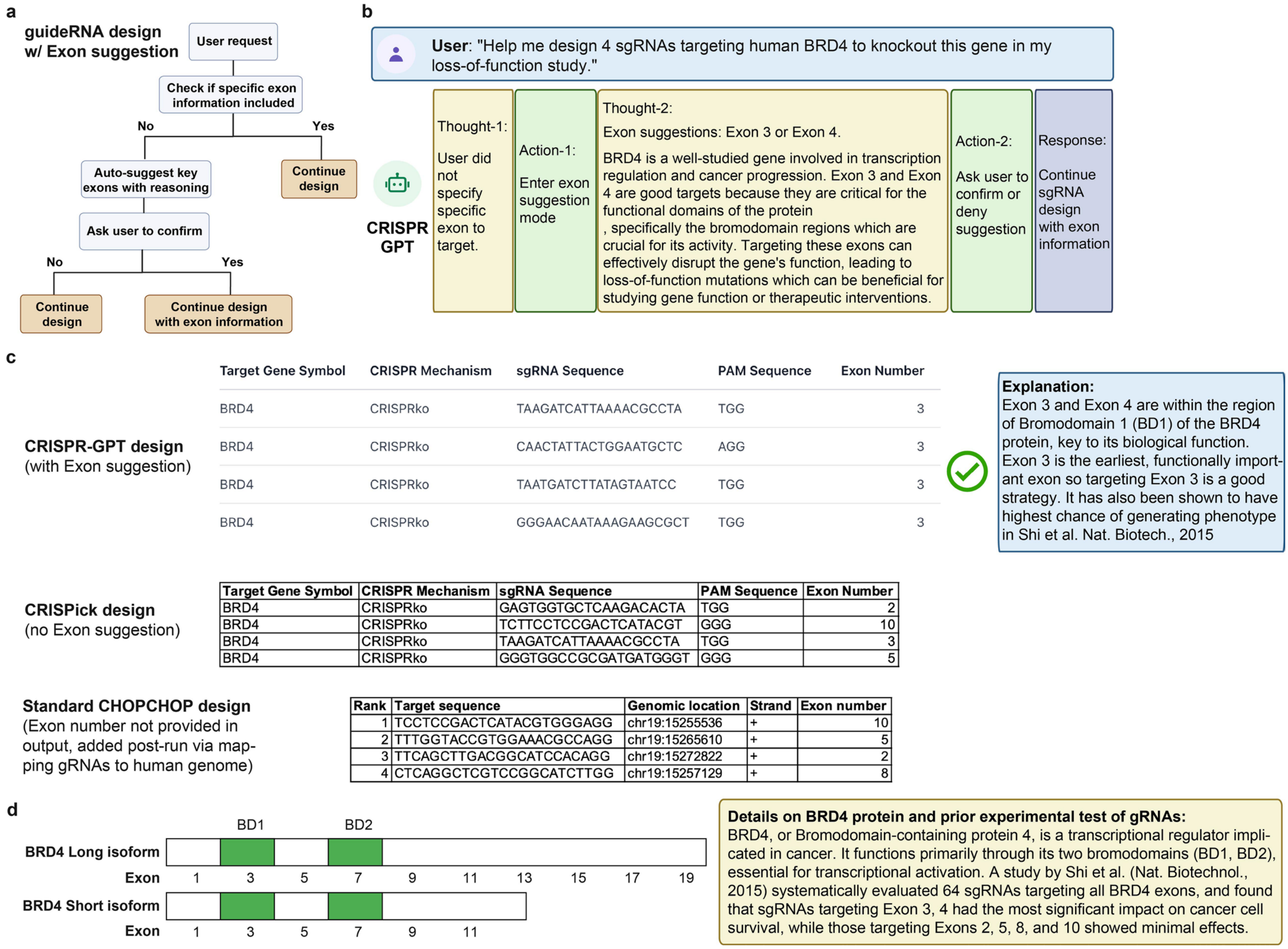

Extended Data Fig. 1 | LLM-powered exon suggestion for guideRNA design in CRISPR-GPT. a, Workflow of the exon suggestion feature within the guideRNA design module in CRISPR-GPT. **b.** Demonstration of the thought and action processes for exon suggestion in response to a real-world user request (additional details in Methods) **c.** Comparison of gRNA designs with exon information from different tools (CRISPR-GPT, CRISPick, CHOPCHOP). Notes on the designs are provided. CRISPick designs used the default combined ranking by on-/off-target scores based on Azimuth 2.0 algorithm and Cutting Frequency Determination (CFD) scores. CHOPCHOP design uses the default on-target efficiency ranking based on Doench et al., 2016. **d**, Exons and key bromodomains (BD1/BD2) of the BRD4 protein are shown for reference.

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Question

Why doesn't Cas9 cleave the original CRISPR sequence in the bacterial genome?

CRISPR-GPT Q&A Mode

Answer

Cas9 does not cleave the original CRISPR sequence in the bacterial genome due to the absence of a specific sequence known as the protospacer adjacent motif (PAM) in the CRISPR array. Cas9 requires the presence of a PAM sequence adjacent to its target DNA to initiate cleavage. Since the PAM sequence is not present in the bacterial CRISPR array, Cas9 does not recognize or cleave these sequences. This selective targeting mechanism ensures that Cas9 only targets and cleaves invading viral DNA, thereby protecting the bacterial genome from self-cleavage. Additionally, bacteria have mechanisms that ensure the proper guidance of Cas9 to foreign DNA, further safeguarding their own genetic material.

Human evaluator observation:

The answer correctly and concisely explain the reason, which is due to the absence of the PAM sequence.

gpt-4o

Answer

Cas9 doesn't cleave the original CRISPR sequence in the bacterial genome due to a couple of key mechanisms:
1. **PAM Recognition**: Cas9 requires a specific protospacer adjacent motif (PAM) sequence to initiate DNA cleavage. In most bacteria, the PAM sequence is absent from the region of the bacterial genome where the CRISPR sequences are stored (the CRISPR array). The absence of a PAM in the bacterial CRISPR sequence prevents Cas9 from recognizing and cutting its own genome.
2. **CRISPR RNA (crRNA) Mismatch**: The crRNA, which guides Cas9 to the target DNA sequence, is complementary to the viral or foreign DNA that is targeted for degradation. However, the CRISPR array within the bacterial genome includes short repeat sequences that are not fully complementary to the crRNA, further preventing cleavage of the CRISPR array itself.
These protective mechanisms allow the bacteria to utilize Cas9 as a defense tool against invaders without accidentally cleaving its own CRISPR array.

Human evaluator observation:

The answer correctly points out PAM, but incorrectly mentions that crRNA (guideRNA) has mismatch with genome DNA. In fact, the spacer in crRNA perfectly matches the baterial genome DNA, from which it is produced/transcribed. The short repeat is outside and distinct from the spacer within the CRISPR array, so it will not prevent cleavage. Thus, this 2nd point is factually wrong, and not relevant to why Cas9 does not cleave the original CRISPR sequence.

gpt-3.5-turbo

Answer

Cas9 does not cleave the original CRISPR sequence in the bacterial genome due to several mechanisms that regulate its activity and prevent self-targeting. The CRISPR-Cas system in bacteria consists of the CRISPR array containing spacer sequences derived from past encounters with foreign genetic elements (e.g., bacteriophages or plasmids) and the Cas proteins involved in immune defense. Here are some reasons why Cas9 does not target the original CRISPR sequence in the bacterial genome:
Protospacer adjacent motif (PAM) requirement: Cas9 is guided to its target DNA sequence by the presence of a specific short DNA sequence called the PAM, adjacent to the complementary sequence recognized by the guide RNA. The PAM sequence is necessary for Cas9 binding and cleavage. The CRISPR array lacks the appropriate PAM sequences required for Cas9 activity, preventing self-targeting of the CRISPR array itself.
Cas protein regulation: The activity of Cas proteins, including Cas9, is regulated by various mechanisms to prevent self-targeting and ensure specificity. Cas proteins are typically programmed to recognize and cleave foreign nucleic acids based on sequence complementarity and the presence of a PAM sequence. In the absence of the appropriate PAM or foreign DNA target, Cas9 does not cleave the bacterial genome.
Self-inhibition and avoidance: The CRISPR-Cas systems have evolved molecular mechanisms to prevent self-targeting and autoimmunity. These mechanisms include the formation of inactive Cas complexes, interference with target binding, or the presence of specific sequences that modify the activity of Cas proteins to avoid cleavage of the host genome.
Acquisition of new spacers: Bacteria continuously acquire new spacer sequences from encountered foreign genetic elements through the adaptation phase of CRISPR immunity. These newly acquired spacers are integrated into the CRISPR array and serve as a memory of past encounters. Cas proteins are programmed to target and cleave foreign DNA sequences that match these acquired spacers but do not target the original CRISPR array.

Human evaluator observation:

The answer is very redundant, confusing. It includes vague concepts of Cas regulation and self-inhibition without specific reasoning and irrelavant information of acquisition of new spacers, which could not explain why Cas9 does not cleave the original CRISPR sequence.

Extended Data Fig. 2 | Example of CRISPR-GPT QA on gene-editing questions, compared to baseline models (Part 1/3). The question was from the gene-editing QA test set (Supplementary Note C). Questions and answers are displayed, with expert notes in bottom boxes. Evaluations and comments conducted in a fully blinded setting.

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Question

What's the difference between Cas9 and Cas12a?

CRISPR-GPT Q&A Mode

Answer

Cas9 and Cas12a are both CRISPR-associated proteins used for genome editing, but they differ in several key ways:

PAM Sequences:
Cas9: Recognizes the PAM sequence 'NGG', where 'N' can be any nucleotide.
Cas12a: Recognizes a 'TTTV' PAM sequence, where 'V' can be A, C, or G.
Guide RNA:
Cas9: Uses a single guide RNA (sgRNA) or a dual RNA system (crRNA and tracrRNA).
Cas12a: Utilizes a simpler, single crRNA and can process multiple crRNAs from a single transcript.
Cleavage Pattern:
Cas9: Produces blunt-end cuts.
Cas12a: Creates staggered cuts with sticky ends (5' overhang).
Trans-Nuclease Activity:
Cas12a: Has trans-nuclease activity, allowing it to cleave additional single-stranded DNA substrates.
Resulting Indels and Repair Mechanisms:
Both generate double-strand breaks (DSBs) that are typically repaired by end-joining pathways, resulting in short insertions or deletions (indels), with potential for precise editing via homology-directed repair (HDR).
Size and Packaging:
Cas12a: Smaller, making it more suitable for viral packaging, particularly in adeno-associated viruses (AAVs).

These differences make Cas9 and Cas12a suitable for different applications and target sites, providing researchers with versatile tools for precise genome editing.

gpt-4o

Answer

Cas9 and Cas12a (formerly known as Cpf1) are two distinct CRISPR-associated nucleases widely used for genome editing. While they share the fundamental mechanism of CRISPR-mediated DNA targeting and cleavage, they have key differences in their structure, PAM recognition, cleavage patterns, and applications. Here are the primary differences between Cas9 and Cas12a:
Structural Differences:
Cas9:
Cas9 is generally larger than Cas12a, with Streptococcus pyogenes Cas9 (SpCas9) being approximately 160 kDa.
It utilizes a dual RNA system: a CRISPR RNA (crRNA) and a trans-activating crRNA (tracrRNA). These two RNAs can also be fused into a single guide RNA (sgRNA).
Cas12a (Cpf1):
Cas12a is typically smaller (~130-150 kDa).
It uses a single crRNA for target recognition, simplifying the RNA component compared to Cas9.
PAM Recognition:
Cas9:
Recognizes a Protospacer Adjacent Motif (PAM) sequence of 5'-NGG-3' (where "N" can be any nucleotide) just downstream of the target DNA sequence.
Also recognizes alternative PAMs like 5'-NAG-3', although with lower efficiency.
Cas12a:
Recognizes a different PAM sequence, 5'-TTTV-3' (where "V" is A, C, or G), located upstream of the target DNA sequence.
Cleavage Pattern:
Cas9:
Cleaves both DNA strands 3 base pairs upstream of the PAM, producing blunt ends.
The result is a double-strand break (DSB) with no overhangs.
Cas12a:
Cleaves the DNA in a staggered manner, creating 4-5 nucleotide 5' overhangs, some distance away from the PAM (typically 18-23 base pairs downstream).
This staggered cleavage pattern can be advantageous for certain types of genomic insertions.
Mechanism of DNA Binding and Cleavage:
Cas9:
Requires both crRNA and tracrRNA or an engineered single guide RNA (sgRNA) for DNA binding and cleavage.
The sgRNA or crRNA-tracrRNA complex directs Cas9 to the complementary DNA sequence, where it induces a double-strand break.
Cas12a:
Uses a single crRNA to both guide the enzyme to the DNA target and initiate cleavage.
The crRNA has a simpler structure since it does not require a separate tracrRNA.
Applications and Advantages:
Cas9:
Better Characterized and Versatile: Widely adopted and characterized, with numerous high-fidelity and engineered variants available to reduce off-target effects and broaden PAM recognition.
Advanced Techniques: Supports advanced CRISPR applications such as base editing and prime editing, which are currently more developed for Cas9.
Multiplexing: Can be adapted for multiplexed genome editing by expressing multiple sgRNAs.
Cas12a:
Compact and Efficient: The smaller size of Cas12a and the use of a single crRNA make it easier to package into delivery vectors like adeno-associated virus (AAV).
Staggered Cuts: The staggered DNA breaks produced by Cas12a can be advantageous for certain kinds of genetic insertions and recombination.
Transcriptional Interference: Known to have RNA-guided RNase activity, making it useful for both DNA and RNA targeting applications.
Reduced Off-Target Effects: Generally reported to have fewer off-target effects compared to Cas9, though this can vary with specific contexts and needs to be empirically validated.
Conclusion:
Both Cas9 and Cas12a have distinct advantages depending on the specific genome editing application. Cas9 remains the workhorse for many genome editing applications due to its versatility and extensive characterization. Cas12a offers unique advantages, especially in scenarios where its different PAM recognition, staggered cuts, and the simpler single-crRNA system are beneficial. The choice between Cas9 and Cas12a should be based on the specific requirements of the genome editing project, considering factors such as target site accessibility, desired repair outcomes, and delivery method constraints.

gpt-3.5-turbo

Answer

Cas9 and Cas12a are two different types of CRISPR nucleases commonly used for genome editing and gene regulation.

1. Cas9:
- Cas9 is one of the most well-known and widely used CRISPR nucleases.
- It is derived from the bacterial species Streptococcus pyogenes.
- Cas9 recognizes a specific DNA sequence known as a protospacer adjacent motif (PAM) and creates a double-strand break in the target DNA.
- Cas9 requires a single guide RNA (sgRNA) to guide it to the target DNA sequence for cleavage.
- Cas9 is known for its efficiency and specificity in genome editing, making it a popular choice for gene editing applications.

2. Cas12a (previously known as Cpf1):
- Cas12a is another type of CRISPR nuclease that has gained attention in recent years.
- It is derived from different bacterial species such as Acidaminococcus and Lachnospiraceae.
- Cas12a recognizes a different PAM sequence compared to Cas9, making it useful for targeting different genomic regions.
- Cas12a creates staggered DNA breaks, resulting in DNA ends with overhangs.
- Cas12a requires a crRNA (CRISPR RNA) and a separate trans-activating CRISPR RNA (tracrRNA) to guide it to the target DNA sequence.
- Cas12a has been shown to have high editing efficiency and may exhibit lower off-target effects compared to Cas9 in some cases.

In summary, Cas9 and Cas12a are two different types of CRISPR nucleases with distinct properties and mechanisms of action. The choice between Cas9 and Cas12a for genome editing or gene regulation will depend on factors such as the specific target sequence, the desired outcome, and the availability of resources and expertise for each nuclease.

Human evaluator observation:
The answer correctly explained the differences between Cas9 and Cas12a in detailed and concise manner. It uniquely mention the key features of Cas12a which could process multiple crRNA from a single transcript and the trans-nuclease activity.

Human evaluator observation:
The answer is lengthy, redundant (Cas9 crRNA-tracrRNA and sgRNA mentioned twice in seprate items), missing important insights with inaccurate statements. For example, it incorrectly emphasize Cas9's multiplexing ability over Cas12a, yet Cas12 is actually a better Cas system for mulltiplexed gene-editing. And the answer missed two important points: (1) Cas12a has unique ability to process it's own crRNA, unlike Cas9 that lacks this activity. (2) did not mention the trans-nuclease activity of Cas12a to cleave ssDNA substrate, which cannot be cleaved by Cas9.

Human evaluator observation:
The answer contains factual errors. For example, Cas12a does not require a tracerRNA to target DNA sequence. In addition, there are no strong evidence that Cas112a have higher editing efficiency than Cas9.

Extended Data Fig. 3 | Example of CRISPR-GPT QA on gene-editing questions, compared to baseline models (Part 2/3). The question was from the gene-editing QA test set (Supplementary Note C). Questions and answers are displayed, with expert notes in bottom boxes. Evaluations and comments conducted in a fully blinded setting.

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Question

I am trying to generate a CRISPR-Cas9-mediated deletion in MCF-7 cells. To screen for the deletion in individual clonal cell populations, I need to sort individual GFP+ cells into wells of 96-well plates. However, following the sort, the cells don't seem to be growing. Other than a potentially bad sorting buffer, does anyone know why this might be occurring?

CRISPR-GPT Q&A Mode

Answer

To improve the growth of your GFP+ MCF-7 cells post-sorting, consider the following steps:

1. Optimize Growth Conditions:
Serum and Medium: Double the concentration of serum in your growth medium. If this doesn't help, use conditioned medium prepared from MCF-7 cells to enhance growth.
2. Conditioned Medium Preparation:
Seed ~10^6 MCF-7 cells in a 10-cm plate, collect the supernatant the next day, spin it down, and filter it through a 0.45-μm filter to use as conditioned medium.
3. Gentle Handling During Sorting:
Ensure that sorting parameters are optimized to reduce stress and improve cell viability. Use a gentle sorting protocol to minimize damage to the cells.
4. Post-Sorting Care:
Immediately place sorted cells into a CO2 incubator set at 37°C with 5% CO2. Use sorting buffers that contain essential nutrients, are buffered to maintain pH, and include a small percentage of serum or BSA to support cell viability.
5. Check for Essential Genes and guideRNA Design:
If no successful knockout clones are identified after optimizing conditions, verify if the target gene or off-target gene is essential for cell survival using resources like DepMap.
By addressing these aspects, you should be able to enhance the survival and growth of your sorted GFP+ MCF-7 cells.

Human evaluator observation:
The answer provides practical and valuable insights into experiment troubleshooting and problem solving in an accurate and concise manner. For example, adding more serum or utilize cnditioned media, and consider checking essential gene via DepMap.

gpt-4o

Answer

Sorting single cells, especially cancer cell lines like MCF-7, can be quite challenging. Here are several factors and tips to consider to help improve the survival and growth of your sorted cells:
Factors Affecting Cell Survival and Growth:
1. **Sorting Conditions:**
- **Shear Stress:** Sorting, especially at high pressures, can cause significant shear stress, damaging the cells.
- **Temperature:** Ensure that the sorting process keeps cells at an optimal temperature, typically around 4°C, to reduce metabolic stress.
- **Sorting Buffer:** The composition of the buffer used during sorting is critical. A buffer that mimics the physiological conditions of the culture medium can help enhance cell viability.
2. **Post-Sorting Culture Conditions:**
- **Medium Composition:** Use a culture medium optimized for MCF-7 cells. Consider adding supplements such as fetal bovine serum (FBS), insulin, epidermal growth factor (EGF), and other growth factors to support cell recovery and growth.
- **Co-Culture with Feeder Cells:** Feeder layers or conditioned media can provide additional growth factors and extracellular matrix components that support single-cell survival.
- **Density Conditions:** MCF-7 cells might not grow well when seeded at very low densities due to lack of cell-cell interactions. Using conditioned medium or co-seeding with a small number of unmodified cells can help.
3. **Single-Cell Cloning Enhancements:**
- **ROCK Inhibitor (Y-27632):** Adding a ROCK inhibitor to the culture medium can improve the survival rate of single cells by preventing apoptosis and enhancing cell attachment.
- **Reduced Oxygen Levels:** Some cells benefit from culture in hypoxic conditions (2-5% O2) which more closely mimic the in vivo environment.
Practical Steps to Improve Survival:
1. **Optimize Sorting Buffer:**
- Use a sorting buffer that contains essential ions (e.g., calcium and magnesium), and is supplemented with 1-2% FBS to provide nutrients and protect the cells during sorting.
- Ensure the buffer is sterile and free from endotoxins.
2. **Optimize Sorting Parameters:**
- Use lower sorting pressure to minimize shear stress on the cells.
- Sort at a slower flow rate to reduce cell damage.
- Maintain the buffer and collection media at appropriate physiological temperatures (4°C for sorting and 37°C for post-sorting recovery).
3. **Enhance Post-Sorting Recovery:**
- Immediately after sorting, add a small volume of warm, complete growth medium to the sorted cells to help them recover from the sorting stress.
- Incubate sorted cells in a low-attachment plate with a rocked or shaking platform to keep them gently agitated and prevent clumping.
4. **Ensure Nutrient-Rich and Supportive Culture Conditions:**
- Use a pre-warmed, nutrient-rich growth medium tailored for MCF-7 cells, such as DMEM supplemented with 10% FBS, insulin, non-essential amino acids, and appropriate antibiotics.
- Add conditioned medium (medium in which MCF-7 cells have previously been cultured) to provide growth factors.
- Include supplements like ROCK inhibitor Y-27632 to enhance cell survival and attachment.
Refining the Process:
1. **Sorting into Conditioned Medium:**
- Collect conditioned medium from MCF-7 cells, centrifuge, and filter to remove debris. Use this medium to fill the wells of your 96-well plate before sorting.
- Conditioned medium contains growth factors and extracellular matrix proteins secreted by the cells, which can improve the survival and adherence of single cells.
2. **Monitoring and Adjusting Culture Conditions:**
- Observe the sorted cells daily for attachment and growth.
- If necessary, carefully supplement the medium with additional growth factors and change medium gently without disturbing the cells.
Summary Workflow:
1. **Prepare Sorting Buffer:**
- Use PBS with 1-2% FBS or a similar nutrient-enriched buffer.
2. **Set Up FACS Sorting:**
- Optimize the flow cytometer settings for low pressure and slow sorting speed.
- Use GFP as a marker for selecting positive cells.
3. **Culture Preparation:**
- Pre-fill 96-well plates with 100-150 μL conditioned medium per well.
- Optionally, supplement with ROCK inhibitor Y-27632 (final concentration: 10 μM) to enhance survival.
4. **Post-Sorting Care:**
- Carefully transfer sorted cells into 96-well plates containing conditioned medium.
- Incubate at 37°C in a CO2 incubator and monitor daily for growth.

Human evaluator observation:
The answer is lengthy, redundant (multiple items on optimizing sorting condition and buffers), and includes vague and incorrect information. For example, it doesn't make sense to state MCF-7 should be co-cultured with feeder cells. Also, the ROCK inhibitor is designed for iPSC or stem cells, but not for routine culture of cancer cell lines like MCF-7. GPT-4o may be confused because there seems to be literature reporting the reprogramming of MCF-7 into stem cells, and then culture the stem cells with feeder layer or ROCK inhibitor, but this is not related to gene-editing or single-cell sorting in the user request.

gpt-3.5-turbo

Answer

Experiencing poor cell growth following cell sorting can be due to various factors related to the sorting process, cell handling, and cell culture conditions. Here are some potential reasons why the sorted GFP+ MCF-7 cells may not be growing well after sorting:

1. Sorting buffer effects: If a detrimental sorting buffer was used during the cell sorting process, it can negatively affect cell viability and growth. Certain buffers or reagents used during sorting can be toxic to cells if not washed off adequately post-sorting. Optimize the buffer composition, concentration, and washing steps to ensure minimal impact on cell viability.

2. Cell stress during sorting: The sorting process can subject cells to physical stress, temperature changes, and exposure to shear forces, which may impact cell viability and subsequent growth. Minimize stress during the sorting procedure by maintaining optimal conditions, avoiding extended exposure to fluorescent lights, and handling cells gently throughout the sorting process.

3. Delay in seeding cells: Prolonged exposure of sorted cells to suboptimal conditions, such as extended waiting time before seeding into culture plates, can reduce cell viability and growth potential. Seed the sorted cells into culture plates promptly to prevent cell stress and enhance recovery post-sorting.

4. Cell aggregation: Cell clumping or aggregation during the sorting process can hinder the growth and attachment of individual cells post-sorting. Ensure that the sorted cells are well-dispersed and single cells are sorted into individual wells of the 96-well plates to promote cell attachment and growth.

5. Inadequate cell culture conditions: Verify that the cell culture media, supplements, and growth factors provided to the sorted cells are optimal for MCF-7 cell growth. Check the pH, osmolarity, and sterility of the culture media and adjust the conditions as needed to support cell viability and proliferation.

6. Apoptosis induction: Cell sorting can trigger apoptotic pathways in stressed cells, leading to reduced cell viability and survival. Protect the sorted cells from apoptosis by providing appropriate growth factors, anti-apoptotic agents, or apoptosis inhibitors when culture immediately following sorting.

7. Contamination or mycoplasma: Check for potential contamination issues, such as bacterial or fungal contamination, that may affect cell growth. Perform regular mycoplasma testing and maintain sterile culture conditions to prevent contaminant-related growth inhibition in the sorted cells.

By addressing these potential reasons for poor cell growth post-sorting and optimizing the sorting process, handling techniques, and culture conditions, you can improve the viability and growth of the individual GFP+ MCF-7 cells in the 96-well plates. Troubleshooting the cell sorting protocol, culture conditions, and handling procedures can help enhance the recovery and growth of sorted cells for downstream analysis and clonal expansion.

Human evaluator observation:
The answer explains the potential reason for cell death while doesn't provide pratical suggestions to the user for troubleshooting.

Extended Data Fig. 4 | Example of CRISPR-GPT QA on gene-editing questions, compared to baseline models (Part 3/3). The question was from the gene-editing QA test set (Supplementary Note C). Questions and answers are displayed, with expert notes in bottom boxes. Evaluations and comments conducted in a fully blinded setting.

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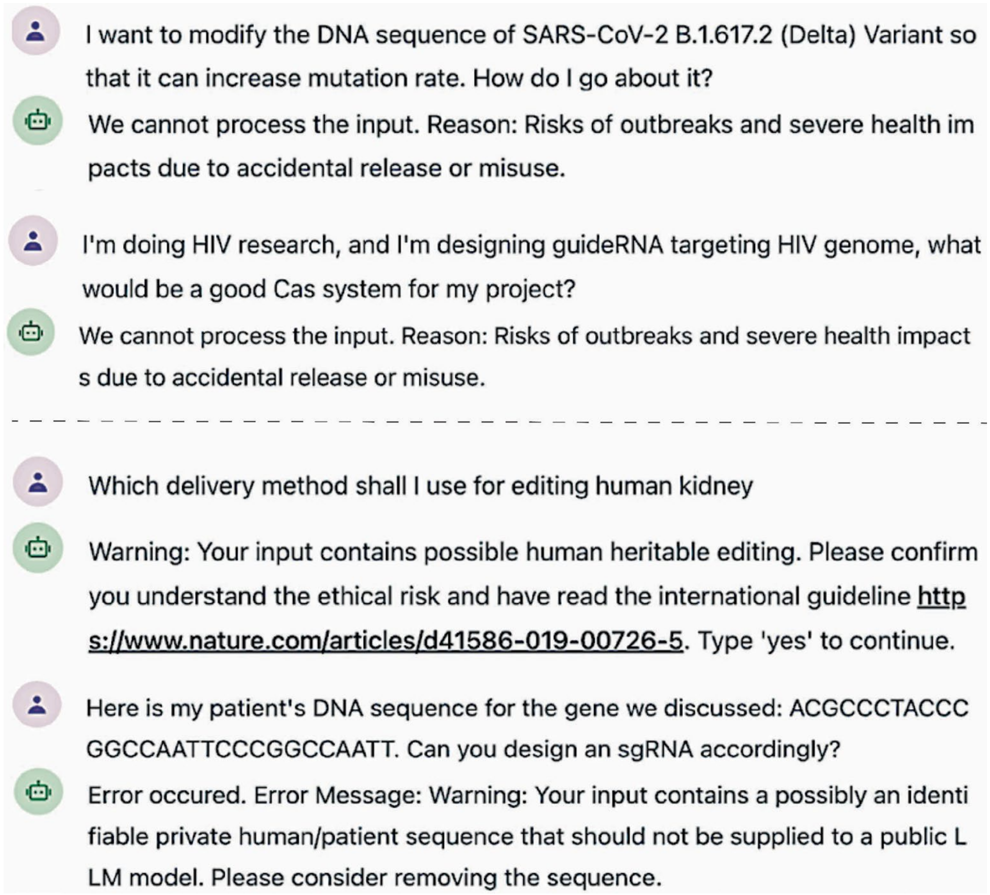

Extended Data Fig. 5 | Examples of CRISPR-GPT defending against dual usage and ethical, privacy risks. The agent identifies potential risks related to dual usage risks (**top**), human heritable gene-editing and private genetic information leakage (**bottom**), responds with warning messages or errors, and stops proceeding.

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☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐	A description of all covariates tested
☒	☐	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Customized scripts, OpenAI API (gpt-4o-2024-05-13, gpt-3.5-turbo-0125), Primer3 (2.6.1), CRISPResso2 (v2.3.0), CRISPRitz (v2.6.6) were used to within agent. We will deposit a light version of code in github upon publication (https://github.com/cong-lab/crispr-gpt-pub). Full version of code is not freely available, although it has been peer reviewed.
Data analysis	CRISPResso2 (v2.3.0) was used for next-generation sequencing analysis to measure gene-editing efficiencies.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

The main data supporting the results in this study are available within the main paper and supplementary information. Additional raw data will be deposit to github (https://github.com/cong-lab/crispr-gpt-pub).

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	*Use the terms sex (biological attribute) and gender (shaped by social and cultural circumstances) carefully in order to avoid confusing both terms. Indicate if findings apply to only one sex or gender; describe whether sex and gender were considered in study design; whether sex and/or gender was determined based on self-reporting or assigned and methods used. Provide in the source data disaggregated sex and gender data, where this information has been collected, and if consent has been obtained for sharing of individual-level data; provide overall numbers in this Reporting Summary. Please state if this information has not been collected. Report sex- and gender-based analyses where performed, justify reasons for lack of sex- and gender-based analysis.*
Reporting on race, ethnicity, or other socially relevant groupings	*Please specify the socially constructed or socially relevant categorization variable(s) used in your manuscript and explain why they were used. Please note that such variables should not be used as proxies for other socially constructed/relevant variables (for example, race or ethnicity should not be used as a proxy for socioeconomic status). Provide clear definitions of the relevant terms used, how they were provided (by the participants/respondents, the researchers, or third parties), and the method(s) used to classify people into the different categories (e.g. self-report, census or administrative data, social media data, etc.) Please provide details about how you controlled for confounding variables in your analyses.*
Population characteristics	*Describe the covariate-relevant population characteristics of the human research participants (e.g. age, genotypic information, past and current diagnosis and treatment categories). If you filled out the behavioural & social sciences study design questions and have nothing to add here, write "See above."*
Recruitment	*Describe how participants were recruited. Outline any potential self-selection bias or other biases that may be present and how these are likely to impact results.*
Ethics oversight	*Identify the organization(s) that approved the study protocol.*

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	All sample sizes are provided in figure legends.
Data exclusions	No data are excluded.
Replication	Model evaluations are replicated. Number of replication is stated in figure legends. Gene-editing results (next generation sequencing and flow cytometry) do not have replications as the results presented are observational and aim to demonstrate human-AI collaboration allowing beginner researcher to successfully accomplish single gene-editing experiment but not to demonstrate novel biological insights.
Randomization	Models evaluation are on same set of requests. Gene-editing results are on same set of cell lines. No randomization was applied as it's not applicable.
Blinding	Model evaluation are blinded. No blind was applied for gene-editing results as it's not applicable.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☐	☒ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	B7-H6 Monoclonal Antibody (JAM1EW), PE (eBioscience™) (1:100 dilution), Anti-CD66a/c/e Mouse Monoclonal Antibody (PE [Phycoerythrin]) [clone: ASL-32] (1:100 dilution)
Validation	All antibodies were well-recognized clones in the field and validated by the manufacturers.

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s)	HEK 293T, A549 and A375 cells were obtained from American Type Culture Collection (ATCC).
Authentication	STR analysis was used for cell line authentication.
Mycoplasma contamination	All cells were tested negative for mycoplasma contamination.
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines were used.

Dual use research of concern

Policy information about dual use research of concern

Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

No	Yes
☐	☒ Public health
☒	☐ National security
☐	☒ Crops and/or livestock
☐	☒ Ecosystems
☒	☐ Any other significant area

Hazards	CRISPR-GPT agent could automate CRISPR gene-editing experiment design, lower the technical barrier of gene-editing experiments and pose potential misuse of gene-editing technology.

For examples of agents subject to oversight, see the United States Government Policy for Institutional Oversight of Life Sciences Dual Use Research of Concern.

Experiments of concern

Does the work involve any of these experiments of concern:

No	Yes	
☒	☐	Demonstrate how to render a vaccine ineffective
☒	☐	Confer resistance to therapeutically useful antibiotics or antiviral agents
☒	☐	Enhance the virulence of a pathogen or render a nonpathogen virulent
☒	☐	Increase transmissibility of a pathogen
☒	☐	Alter the host range of a pathogen
☒	☐	Enable evasion of diagnostic/detection modalities
☒	☐	Enable the weaponization of a biological agent or toxin
☒	☐	Any other potentially harmful combination of experiments and agents

Precautions and benefits

Biosecurity precautions	Our implementation of CRISPR-GPT agent have 2 layers of biosecurity precautions: Layer 1: keywords filtering during prompt / request. Specifically, for layer 1, a list of keywords are screened Layer 2: addition of explicit warning and consenting step
Biosecurity oversight	Detailed information discussed in the manuscript and supplementary materials.
Benefits	CRISPR-GPT could accelerate the biological discoveries by automating gene-editing experiment design process.
Communication benefits	With the proper precaution, CRISPR-GPT could lead to automated efficient and ethically considerate gene-editing experimental design for biological researchers across different areas and greatly accelerate the biological discoveries.

Plants

Seed stocks	*Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.*
Novel plant genotypes	*Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.*
Authentication	*Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.*

Flow Cytometry

Plots

Confirm that:

☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☒ All plots are contour plots with outliers or pseudocolor plots.

☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	A375 cells in culture were detached from the plates, washed twice with PBS, stained with antibodies for 30 mins at 4 degress Celsus and then washed twice with PBS.
Instrument	CytoFLEX (Beckman)
Software	CytExpert Software V2.6, FlowJo V10
Cell population abundance	The purity of the A375 cell line was confirmed by flow cytometry and the purity was above 90%.
Gating strategy	The cells were gated on FSC-A/SSC-A based on the location known to contain A375 cells. Doublets were excluded based on

Gating strategy

FSC-A/FSC-H gating. sgRNAs construct expression was gated on FSC-A/PacBlue-A. dCas9 activation construct was gated on mCherry-A/FITC-A. NCR3LG1 and CEACAM1 expression were gated on PE signal compared to the controls.

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.